Consumption-Based Money Demand: Theoretical Properties and Transaction-Level Evidence*

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Abstract

How much do monetary frictions matter in modern economies? We introduce an endogenous consumption-saving decision into a continuous-time Baumol–Tobin model in which households use money to consume and pay a fixed cost to rebalance between money and illiquid assets. The model is tractable enough to analytically prove a variety of results. First, we show that the marginal propensity to consume (MPC) exhibits several properties observed in the data: MPCs are large, including among wealthy households, and decline with liquidity, while intertemporal MPCs decay slowly. Next, we show that the cashless limit is not cashless: as the adjustment cost vanishes, MPC distortions persist and can even intensify. Because of the Baumol-Tobin friction, MPCs aggregate exactly, and distortions disappear only under the Friedman rule. At the aggregate level, fixed costs make money demand state-dependent and its response to monetary shocks persistent. We characterize this sluggish response using a functional sufficient statistic based on steady-state conditional adjustment functions. Then, we use transaction-level account data to document adjustment behavior in the US, calibrate the model, and estimate the functional sufficient statistic.

JEL Codes: E41, E21, D14

Keywords: Money demand; marginal propensity to consume; liquidity; adjustment costs; sufficient statistics; transaction-level data

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1 Introduction

Do monetary frictions matter for the macroeconomy? In the dominant New Keynesian approach (Woodford, 2003), literal money does not appear; "monetary policy" is interest rate policy. Is this a reasonable approximation for modern economies? In this paper, we argue that it is not.

We draw conclusions from a streamlined theory of consumption-based money demand. Households must use money to consume, but face a Baumol-Tobin friction: they must pay a fixed cost of adjustment to rebalance their portfolio of money and illiquid assets. In these types of models, expenditures are typically assumed to be exogenous (e.g., Miller and Orr 1966; Constantinides 1976; Frenkel and Jovanovic 1980; Attanasio et al. 2002; Alvarez and Lippi 2009). Our modification is to give households a standard consumption-savings decision. Because our setting is both relatively simple and in continuous time, the model is highly tractable and we can characterize many theoretical properties.

We begin by showing that the model is consistent with observed patterns in the marginal propensities to consume (MPC). MPCs are large even for high-income households (Kaplan et al., 2014; Kueng, 2018) and, they have little correlation with net wealth when controlling for portfolio composition (Fagereng et al., 2021; Crawley and Kuchler, 2023; Bilbiie et al., 2025).[1] This occurs in the model because the consumer's marginal asset is money, which pays a poor return. Next, MPCs are higher for low-liquidity households (Souleles, 1999; Broda and Parker, 2014; Parker and Souleles, 2019; Jappelli and Pistaferri, 2020; Fagereng et al., 2021; Gelman, 2021; Aydin, 2022; Ganong et al., 2024); when agents have small money balances in the model, they are willing to reduce consumption in order to defer paying the fixed cost, which raises the MPC. Finally, the slope of the intertemporal MPC (iMPC) is small (Johnson et al., 2006; Parker et al., 2013; Kueng, 2018; Fagereng et al., 2021; Aydin, 2022; Auclert et al., 2024); again, consumption smoothing causes this in the model where the poor return on money causes household consumption to fall between adjustments.

The theory reveals a surprising new result: *the cashless limit is not cashless.* As the fixed cost becomes small, the MPC distortion does not go to zero. In fact, the MPC becomes more distorted. Thus, the limiting case is not equivalent to the frictionless model. Even in a modern high-velocity economy, the monetary friction cannot be ignored.[2]

We prove several more surprising features of consumption behavior in the model. First, the

[1] This suggests that borrowing constraints are insufficient to explain large MPCs, as Aiyagari-Bewley-Huggett models predict MPCs are much larger for low-wealth individuals. Binding constraints are the reason MPCs do not aggregate in these models.

[2] This result is thematically related to Lagos and Zhang (2022) and Lagos (2026), which also show that a cashless limit need not be equivalent to an economy without money, although the mechanism is different. In their models, the option to settle transactions with money constrains credit intermediaries' market power even as the use of money vanishes.

MPC exhibits perfect aggregation. This is not a trivial property; other functions do not necessarily aggregate. The MPC aggregates because of the endogenous portfolio adjustment; in contrast, models that do not allow households to pay the fixed cost and adjust at any time do not feature aggregation (e.g., Kaplan and Violante 2014, 2022). We also show this result is a fundamental feature of Baumol-Tobin adjustment costs, and is more general than our specific setting. Perfect aggregation is a desirable property because recent empirical work by Bilbiie et al. (2025) has shown using Norwegian administrative data that the average and aggregate MPC correspond very closely. Second, risk is not a necessary determinant of the MPC distortion; we show that income and expenditure risk raise the MPC, but the MPC is distorted even in the riskless case. Third, the iMPC slope is not policy invariant. The slope is steeper if the inflation rate is higher. Finally, all of the MPC distortions disappear only under the Friedman rule, i.e., there is sufficient deflation such that the real return on money matches that of illiquid assets.

Next, we evaluate the consequences for aggregate money demand and emphasize one main conclusion: money demand is state-dependent. Factors that traditionally affect household money demand, such as cash infusions, interest rate changes, or inflation, only do so slowly over time. This is because these factors affect the rate of portfolio adjustment, but do not immediately change the behavior of households deep in the inaction region.[3] The result that money demand is state-dependent contrasts with common simplifications, such as putting money in the utility function, where households can change their real balances immediately.

To quantify the sluggishness of money demand, we derive a *functional sufficient statistic* in a generalized model. This result gives the impulse response function (IRF) of money demand with respect to a monetary shock, in terms of functions that are measurable in the steady-state distribution. Specifically, the IRF can be written in terms of steady-state conditional functions that describe the timing and size of households' portfolio adjustments. Moreover, this functional sufficient statistic applies to a far more general setting than our benchmark model.

To evaluate the theory, we turn to transaction-level checking account data in the US. We use anonymous individual-account-by-transaction-level data from Fiserv to measure the liquidity adjustment behavior of US households. Fiserv is a leading provider of payments and financial services technology solutions. One of its key businesses involves providing financial technology solutions to banks and credit unions, helping them maintain customer deposit and loan accounts. Our dataset includes transaction-level data from 64 banks and credit unions, covering more than 25 million deposit accounts. We use the data to first describe liquidity adjustment behavior in the US, documenting a number of moments describing checking account transactions, which we use to calibrate the tractable

[3]This mechanism is similar to the sluggish response found in Alvarez et al. (2009), a cash-in-advance model where households are only able to liquidate every N periods.

model. And second, we calculate the functional sufficient statistic and find that money demand is quantitatively sluggish; with a fixed interest rate, deviation in real balances after a monetary shock has a half-life of approximately 60 days.

Our model joins a voluminous literature that follows Baumol (1952) and Tobin (1956). Canonical work with risk in continuous time includes Constantinides (1976) and Frenkel and Jovanovic (1980). Modern treatments introduce stochastic low-cost withdrawal opportunities and estimate the model using household data (Alvarez and Lippi, 2009), allow for uncertain lumpy expenditures (Alvarez and Lippi, 2013), and endogenize the choice between cash and credit (Alvarez and Lippi, 2017). Alvarez et al. (2019) study the welfare implications of inventory-theoretic money demand.

Early progress combining monetary frictions with endogenous consumption began with the cash-in-advance literature (Lucas Jr., 1982; Lars E. O. Svensson, 1985; Lucas and Stokey, 1987). Several papers studied the microeconomic optimization problem for endogenous consumption with Baumol-Tobin costs in discrete time, including Corbae (1993) and Li (2009); Silva (2012) studies the problem in continuous time and shows that endogenous consumption and money management raise the welfare cost of inflation. Another approach that simplifies general equilibrium analysis is to combine Baumol-Tobin costs with complete markets, as in Alvarez et al. (2002) or Khan and Thomas (2015). More recent work has focused on two-asset models where the illiquid asset is also subject to frictions, usually a borrowing constraint. Most of these models are in discrete time; Ragot (2014) is the closest, with only borrowing and cash-in-advance constraints in addition to the Baumol-Tobin costs. In many cases the Baumol-Tobin costs are modified so that model behavior better matches the data, e.g., Kaplan et al. (2018) assume additional non-linearity in the cost of adjustment, and Bayer et al. (2019) assume the costs are stochastic. In other cases households are only allowed to adjust at random opportunities (Luetticke, 2021; Kaplan and Violante, 2022; Di Tella et al., 2024).

The remainder of the paper is organized as follows. Section 2 introduces the household problem. Section 3 characterizes consumption and portfolio adjustment in a deterministic benchmark, including the cashless limit and intertemporal MPCs, and Section 4 extends the analysis to income risk. Section 5 aggregates household behavior and studies the dynamics of money demand. Section 6 derives the functional sufficient statistic. Section 7 describes the transaction-level evidence and calibration, and Section 8 concludes.

2 The Model

This section lays out our model of consumption-based money demand, focusing on the microeconomic decisions. We delay our description of aggregation and the macroeconomy until Section 5.

2.1 Households

Time is continuous and there is a unit measure of households. Households enjoy utility $u(C)$ from consumption C, discounted at rate $\rho > 0$.

They can hold two types of wealth: money M and illiquid assets A. They are both denominated in units of consumption so M denotes real balances. Money is liquid and can be exchanged for consumption at no cost. Asset A is illiquid; the household must pay the fixed utility cost $\varkappa$ to change its holdings of A.

The household earns income rA. r is the real return, and A is the household's illiquid wealth, which may include non-financial assets such as human capital. Some of the income ηA is paid in money (such as wages and dividends) while αA is paid in illiquid assets (such as capital gains or implicit rent from owner-occupied housing), with $\alpha = r - \eta$. Parameters r, α, and η are all constant and positive. The household must respect the natural borrowing limit $A \geq 0$.

The household spends money on consumption C, which it chooses, as well as a stochastic term q which is exogenous:

$$q = s(A)dW^q \tag{1}$$

where dW^q is an exogenous Brownian motion, and $s(A)$ is a function that controls the standard deviation of the process. It is allowed to depend on the illiquid asset A, so that q can represent risky financial returns, human capital returns, and costs associated with home ownership or other durable consumption goods.

Because of the fixed cost to adjusting illiquid asset holdings, household behavior is characterized by an *inaction region*. Inside this region, illiquid assets simply accumulate, depending on the illiquid share α of income:

$$dA = \alpha A dt \tag{2}$$

while the real money balance follows

$$dM = (\eta A - \pi M - C)dt - q \tag{3}$$

where $\pi = \frac{d \log P}{dt}$ is the inflation rate. Therefore, in the inaction region the Hamilton-Jacobi-Bellman (HJB) equation is

$$\rho V(M, A) = \max_C u(C) + V_M(M, A)(\eta A - \pi M - C) + V_A(M, A)\alpha A + V_{MM}(M, A)\frac{s(A)^2}{2} \tag{4}$$

2.2 Optimality Conditions

Optimal household behavior is characterized by the following conditions. Consumption satisfies the first-order condition

$$u'(C) = V_M(M, A) \tag{5}$$

After paying the fixed cost, the household allocates its assets optimally, selecting a "reset" money balance M^* and asset level A^*. These choices must satisfy the optimal reset condition:

$$V_M(M^*, A^*) = V_A(M^*, A^*) \tag{6}$$

When a household's money balance hits zero, it must pay the utility cost $\varkappa$ to liquidate assets. It will reset to an optimal mix (M^*, A^*), which must satisfy the portfolio constraint

$$A = A^* + M^* \tag{7}$$

This gives the first value-matching condition:

$$V(0, A) = V(M^*, A^*) - \varkappa \tag{8}$$

Thus the $M = 0$ line is the lower boundary of the inaction region. There is also an upper boundary. For a money and asset position $(\bar{M}, \bar{A})$ on this upper boundary, the value-matching condition is

$$V(\bar{M}, \bar{A}) = V(M^*, A^*) - \varkappa \tag{9}$$

where now the reset allocation must satisfy the portfolio constraint

$$\bar{M} + \bar{A} = A^* + M^* \tag{10}$$

For $\sigma > 0$, smooth-pasting holds at the upper boundary: the marginal value of money equals the marginal value of assets,

$$V_M(\bar{M}, \bar{A}) = V_A(\bar{M}, \bar{A}) \tag{11}$$

The normalized form of this condition is given in Proposition 1.

Because the household has a two-dimensional state space, it has a two-dimensional inaction region. The next section shows that under homotheticity assumptions, this problem simplifies.

2.3 Normalization

The household's decision problem has two state variables: money M and assets A. For simplicity, it is helpful to reduce the problem to a single state variable. To do so, we assume that $s(A) = \sigma A$, so that the standard deviation of the random income component scales with assets. Then, we rescale the variables in the HJB by assets, so that $m \equiv \frac{M}{A}$ represents the money-to-asset ratio and $c \equiv \frac{C}{A}$ represents the consumption-to-asset ratio. Lastly, we assume that agents have log utility $u(C) = \log C$.[4]

Proposition 1 simplifies the problem by separating the HJB into components that depend on illiquid assets A and the normalized money balance $m = \frac{M}{A}$. It introduces a composite parameter ι defined as

$$\iota \equiv i - \eta = \pi + r - \eta = \pi + \alpha$$

When all income is paid to the illiquid account ($\eta = 0$), then ι is simply the nominal interest rate i, and i is related to inflation π by $i = \pi + r$.

Proposition 1 *If households have log utility and $s(A) = \sigma A$, then the HJB can be rewritten in terms of normalized variables by*

$$V(M, A) = v(m) + w(A)$$

where

$$\rho v(m) = \max_c \log c + v'(m)(\eta - \iota m - c) + v''(m)\frac{\sigma^2}{2} \tag{12}$$

$$\rho w(A) = \log A + \alpha A w'(A) \tag{13}$$

Normalized money follows the diffusion

$$dm = (\eta - \iota m - c)dt - \sigma dW^q \tag{14}$$

Letting c^* and m^* denote the reset values, the optimal reset condition is

$$c^* = \rho(m^* + 1) \tag{15}$$

and the value-matching conditions are

$$v(m^*) - v(0) = \frac{1}{\rho}\log(m^* + 1) + \varkappa \tag{16}$$

[4] Appendix C extends the model to power utility with an adjustment cost that preserves this normalization.

$$v(m^*) - v(\bar{m}) = \frac{1}{\rho}\log(1+m^*) - \frac{1}{\rho}\log(1+\bar{m}) + \varkappa \tag{17}$$

For $\sigma > 0$, smooth-pasting holds at the upper boundary:

$$\bar{c} = \rho(1+\bar{m}) \tag{18}$$

Proof: Appendix A.1.

In the normalized model, there is a single state variable: normalized money m. We write the normalized consumption function as $c(m)$. Lemma 1 shows that this normalized consumption function is sufficient for analyzing the MPC:

Lemma 1 *For the consumption function $C(M, A)$, the marginal propensity to consume (MPC) out of money $C_M(M, A)$ is given by the normalized MPC:*

$$C_M(M, A) = c'(m)$$

Proof: Appendix A.2.

Section 4 continues with this model, presenting the main theorem containing the general model properties. But first, to build intuition, the next section explores the main model properties in a simple, tractable case.

3 A Deterministic Case

The numerical results in this section set $\eta = 0$ and hold the other parameters at the baseline values in Table 2, except when explicitly varying the adjustment cost.

To develop understanding, we first solve the household's problem for the special case without risk and with only illiquid income ($\sigma = \eta = 0$). This case suffices to capture the forces and intuition that matter for our main conclusions.

3.1 Household Behavior

The special case with all illiquid income ($\eta = 0$) is particularly tractable; we will relax this assumption in Section 4.2. The following proposition collects the main features of household behavior in this special case.

Proposition 2 *If $\sigma = \eta = 0$ and $i > 0$, normalized consumption and money evolve after a reset by*

$$c(t) = c^* e^{-(\rho+i)t} \qquad m(t) = \frac{c(t)}{\rho} - \left(\frac{c(t)}{c^*}\right)^{\frac{i}{\rho+i}}$$

where $t \in [0, T]$ is time since the most recent reset. The inverse consumption function and its derivative are

$$m(c) = \frac{c}{\rho} - \left(\frac{c}{c^*}\right)^{\frac{i}{\rho+i}} \qquad m'(c) = \frac{1}{\rho} - \frac{i}{\rho+i}\frac{c^{-\rho/(\rho+i)}}{(c^*)^{i/(\rho+i)}}$$

The time between resets is

$$T = \frac{1}{\rho}\log(1+m^*)$$

and the optimal reset values satisfy

$$c^* = \rho(1+m^*)$$

The adjustment cost and reset value satisfy

$$\varkappa = \frac{i}{\rho^2}\left[\log(1+m^*) - \frac{m^*}{1+m^*}\right]$$

where the right-hand side is strictly increasing in m^*. The MPC $c'(m)$ exceeds ρ and is strictly decreasing in m.

Proof: Appendix A.4.

Proposition 2 makes clear that the model features nominal non-neutrality: the nominal interest rate $i = \pi + r$, and therefore inflation π, distort the paths of money and consumption. In levels, consumption between resets evolves according to

$$C(t) = C(0)e^{-(\rho+\pi)t}$$

Thus inflation has real effects even in this deterministic case.

At the Friedman rule $i = 0$, money and illiquid assets have the same return and the household does not choose a finite reset balance. The consumption policy is $c(m) = \rho(1+m)$, so $c'(m) = \rho$.

3.2 The MPC out of Money

When all income is illiquid, Corollary 1 shows that the MPC at the zero-balance boundary is simply $\rho + i$. This is a clear demonstration that low-liquidity individuals have high MPCs, a consistent pattern in the studies of consumption (Souleles, 1999; Broda and Parker, 2014; Parker and Souleles, 2019; Jappelli and Pistaferri, 2020; Fagereng et al., 2021; Gelman, 2021; Aydin, 2022; Ganong et al.,

2024). Moreover, the distortion is increasing in the nominal interest rate and disappears under the Friedman rule ($i = 0$).

Corollary 1 *If $\sigma = \eta = 0$ and $i > 0$, the marginal propensity to consume at the boundary is*

$$C_M(0, A) = \rho + i$$

Proof. Because there is no risk, the MPC satisfies

$$c'(m) = \frac{\dot{c}}{\dot{m}} = \frac{-(\rho + i)c}{-im - c}$$

which at the $m = 0$ boundary becomes

$$c'(0) = \rho + i$$

■

The proof of Corollary 1 is the clearest illustration of how the monetary friction raises the MPC, unless $i = 0$. This result is caused by *consumption smoothing*, as it follows mathematically from the Euler equation and law of motion for money. Moreover, the result also shows why the distortion will not disappear as the monetary friction goes to zero: the proof does not depend on the adjustment cost. The MPC at the boundary is $\rho + i$, no matter the size of the inaction region.

Emerging empirical evidence suggests that while MPCs are correlated with liquidity, even high-liquidity individuals have elevated MPCs (Holm et al., 2021; Boehm et al., 2025; Bilbiie et al., 2025). Corollary 2 demonstrates that this too is a feature of the model. The MPC at the reset point is above the frictionless MPC whenever $i > 0$. Because m decreases between resets and the MPC is decreasing in m, all individuals have elevated MPCs relative to the frictionless benchmark.

Corollary 2 *If $\sigma = \eta = 0$ and $i > 0$, the marginal propensity to consume at the reset point is*

$$C_M(M^*, A) = \frac{\rho(\rho + i)}{\rho + is^*}$$

where $s^* \equiv \frac{m^*}{m^*+1}$ is the liquid share of wealth at the reset. This exceeds the frictionless MPC ρ whenever $i > 0$.

Proof: Appendix A.5.

Corollaries 2 and 1 imply that the MPC curve is relatively flat. Because $c'(m)$ is a decreasing

function, $C_M(0, A)$ is the maximum and $C_M(M^*, A)$ is the minimum. Their ratio is

$$\frac{C_M(0, A)}{C_M(M^*, A)} = \frac{1 + \frac{i}{\rho} s^*}{1}$$

In the US, the nominal interest rate i is low, as is the liquid wealth share s^*. As a result, the increase in the MPC from the reset point to the boundary is close to zero in the model where all income is paid to the illiquid account. Figure 1 demonstrates this by plotting the MPC $c'(m)$ against the normalized money balance. The MPC rises somewhat as balances approach zero, but the dominant distortion is that for all m levels, the MPC is substantially larger than the frictionless MPC. And Corollary 1 implies that at the boundary, this difference is exactly the nominal interest rate i.

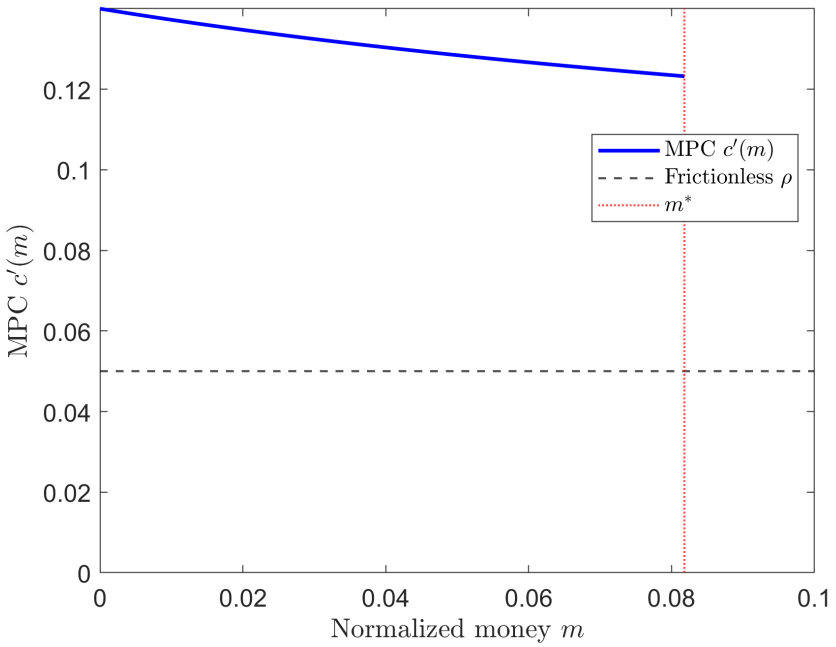


Figure 1: MPC as a Function of Normalized Money

Note: The MPC out of money, $c'(m)$, as a function of normalized money $m \in [0, m^*]$. The MPC is strictly decreasing in m and strictly exceeds the frictionless benchmark ρ (dashed) for all $m < m^*$ whenever $i > 0$ (Proposition 2). The vertical dotted line marks the reset point m^*.

MPCs at the critical points are useful for a back-of-the-envelope calculation of the MPC in the economy, as they bound the range of MPCs on $[0, m^*]$. In order to be more exact, Corollary 3 gives the average MPC experienced over time between resets.

Corollary 3 *If $\sigma = \eta = 0$ and $i > 0$, the average MPC is*

$$\overline{MPC} = \rho \frac{\log\left(1 + \frac{\rho+i}{\rho} m^*\right)}{\log(1 + m^*)}$$

Proof: Appendix A.6.

3.3 The Cashless Limit is not Cashless

This section describes a surprising result: the limit of this economy as the adjustment cost goes to zero is *not* equivalent to the frictionless economy. The MPC distortion caused by the monetary friction does not disappear for arbitrarily small costs. For this reason, we say the *cashless limit is not cashless.*

What happens to the average MPC as $m^* \to 0$? Both the numerator and denominator in Corollary 3 go to zero. The endpoints both converge to $\rho + i$; the average will as well, as Proposition 3 shows.[5]

Proposition 3 (The Cashless Limit) *If $\sigma = \eta = 0$ and $i > 0$, then*

$$\lim_{\varkappa \to 0} \overline{MPC} = \lim_{m^* \to 0} \overline{MPC} = \rho + i > \rho$$

Proof: Appendix A.7.

It may be surprising that Corollary 3 implies the average MPC $\overline{MPC}$ is *decreasing* in m^*. Figure 2 demonstrates this by varying the adjustment cost $\varkappa$ and plotting the average MPC. Increasing the utility cost $\varkappa$ increases the reset value m^*. When the adjustment cost is low, so that the reset liquidity is also low, the average MPC is substantially larger than the frictionless MPC. Moreover, the average MPC only approaches the frictionless MPC very slowly; adjustment costs need to become extreme so that households hold very large liquid balances and rarely reset in order to get smaller MPCs. This is because the MPC at the boundary remains fixed at $\rho + i$, and Corollary 2 implies that the reset liquidity share $s^* = \frac{m^*}{m^*+1}$ needs to get close to 1 for the reset MPC to approach the frictionless ρ value.

This result is remarkable because it implies that *the frictionless economy is not the cashless limit.* When the friction $\varkappa$ goes to zero, the maximum money balance m^* goes to zero, and the economy's monetary velocity approaches infinity. And yet, the average MPC remains distorted. Indeed, it becomes *the most* distorted.

[5]The positive limiting MPC distortion also holds with a more general utility function, as Appendix C shows.

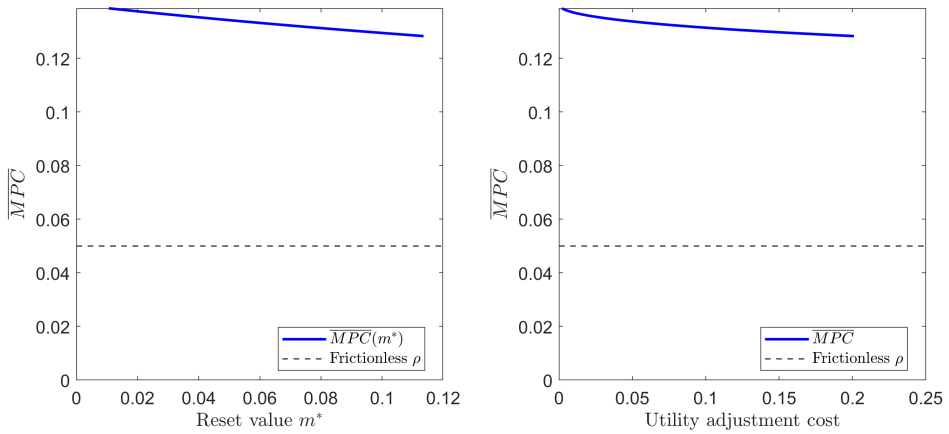


Figure 2: Average MPC as a Function of the Adjustment Cost and Reset Value

Note: Left panel: the average MPC against the reset value m^*. Right panel: the average MPC against the utility adjustment cost $\varkappa$. Each point corresponds to a different value of $\varkappa$; both axes are equivalent parameterizations of the same object. The average MPC is decreasing in m^* (equivalently, decreasing in $\varkappa$): smaller reset buffers amplify the MPC distortion. The dashed line is the frictionless MPC ρ.

Only under the Friedman rule does the average MPC distortion disappear. When $i = 0$, the MPC is $c'(m) = \rho$ everywhere.

How is it possible that the average MPC is so large even when the monetary friction is small? The key is that the MPC is large when $m > 0$, but consumption behaves differently at the boundary. When monetary balances hit zero, households liquidate, and consumption jumps discretely. This gives consumption a "sawtooth" pattern (Figure 3, left panel). It is easy to see why this arises: balances in money demand models commonly follow sawtooth patterns, where the balance declines to zero, then jumps to a reset point before continuing the pattern. The normalized consumption function $c(m)$ is increasing, so consumption must also follow a sawtooth pattern. However, unlike Baumol (1952), the sawtooths are no longer linear; money balances slow their decline as households reduce their rate of consumption before a reset (Figure 3, right panel).

3.4 Intertemporal MPCs

The impact MPC is the horizon-zero effect of a monetary increase. It is also useful to ask how a small increase in current money affects consumption at later horizons. For a household with current assets A_0 and normalized money m, define

$$\mathcal{U}(m,h) \equiv \frac{C_h}{A_0} \qquad \mathcal{I}(m,h) \equiv \partial_m \mathcal{U}(m,h)$$

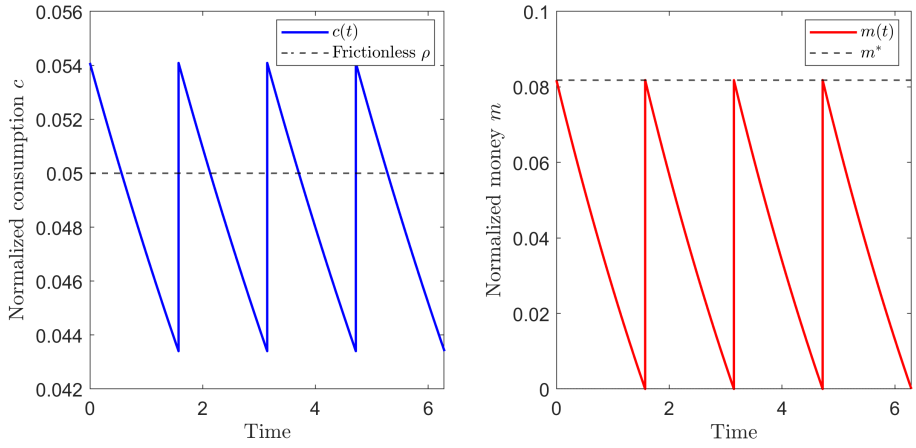


Figure 3: Sawtooth Dynamics of Consumption and Money

Note: Normalized consumption $c(t) = C(t)/A(t)$ (left) and normalized money $m(t) = M(t)/A(t)$ (right) over four reset cycles. At each reset, the household pays the fixed cost $\varkappa$ and restores money to m^*. Between resets, consumption declines at rate $\rho + i$ and money drains to zero, after which the household must reset again. The dashed line in the left panel is the frictionless consumption ratio ρ; the dashed line in the right panel marks m^*.

Since $M_0 = A_0 m$, $\mathcal{I}(m, h)$ is the marginal effect of a real balance increase on consumption at horizon h.

Proposition 4 (Riskless iMPC Horizon Elasticity) *If $\sigma = \eta = 0$ and $i > 0$, then before the household's next reset the iMPC is*

$$\mathcal{I}(m, h) = e^{-(\rho+\pi)h} c'(m) \tag{19}$$

Thus except at reset times, the horizon elasticity of the iMPC is constant:

$$\partial_h \log \mathcal{I}(m, h) = -(\rho + \pi) \tag{20}$$

If $\rho = r$, then the elasticity is simply $\partial_h \log \mathcal{I}(m, h) = -i$.

Proof: Appendix A.8.

At reset horizons the pointwise response has one-sided jumps because the baseline household resets discretely, so the marginal transfer changes the timing of that reset. Figure 4 plots these horizon profiles across different monetary frictions, with each household starting from its own reset target m^*. Consistent with Figure 2, the impact MPC (at the $h = 0$ intercept) rises as the monetary friction κ shrinks. But it is not just the impact MPC; the entire iMPC curve increases as the friction goes to zero. Of course, this increase is not uniform; adjustments become more frequent as the cost decreases, so the sawtooths become smaller and more common. The cashless limit is not cashless at

any horizon: the frictionless iMPC is a horizontal line at ρ, far lower than the iMPC corresponding to the smallest friction.

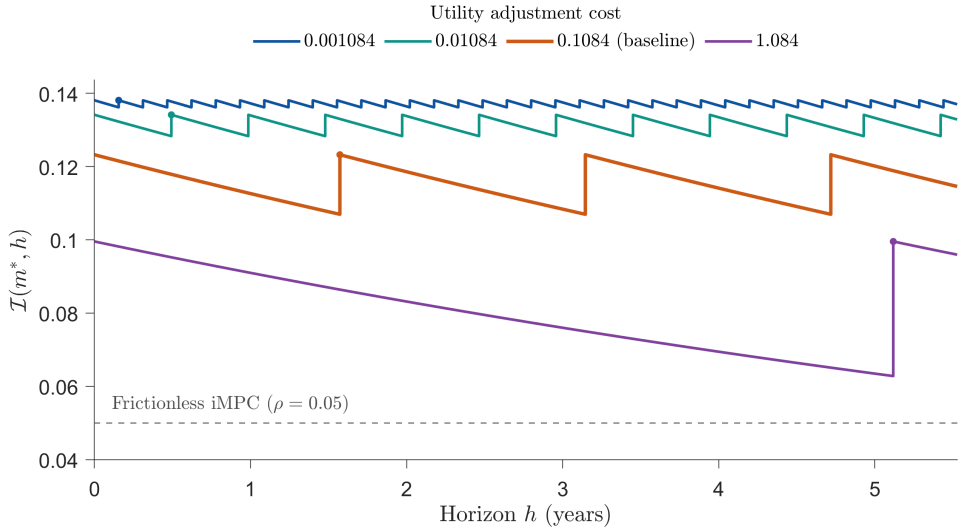


Figure 4: Riskless Intertemporal MPCs

Note: The individual iMPC $\mathcal{I}(m^*, h)$ for households at the optimal reset m^* under different adjustment costs. The utility adjustment costs are 0.01, 0.1, 1, and 10 times the baseline $\varkappa$, holding the other parameters fixed. The legend reports the cost values, with the baseline curve drawn more heavily. Horizons are measured in years. Filled circles mark each curve's first deterministic reset. Between reset dates, the iMPC obeys the no-reset formula (19); at reset dates, the pointwise response has one-sided jumps because a marginal money transfer shifts the reset time.

4 Income Risk

This section returns to the general model. Section 4.1 characterizes the riskless model with non-zero liquid income; this is a necessary input to Section 4.2's analysis of the model with non-zero risk.

4.1 The Riskless Benchmark

Section 3 set $\eta = 0$; this section derives the equilibrium behavior when $\eta \geq 0$.

Let c_0 denote its inaction-branch consumption function and let m_0^* and $\bar{m}_0$ denote its reset value and upper boundary. Write the one-sided boundary values as

$$c_T \equiv \lim_{m \downarrow 0} c_0(m) \qquad \bar{c}_0 \equiv \lim_{m \uparrow \bar{m}_0} c_0(m)$$

The riskless drift is $\mu_0(m) \equiv \eta - \iota m - c_0(m)$. We maintain $i > 0$, $\iota > 0$, $c_T > \eta$, and finite thresholds $0 < m_0^* < \bar{m}_0$ throughout this section.

Lemma 2 *The riskless inaction-branch consumption function is uniquely determined by m_0^*. Its derivatives satisfy*

$$c_0'(m) > \rho \qquad c_0''(m) < 0$$

for $m \in [0, \bar{m}_0)$. Moreover,

$$\bar{c}_0 > \rho(1 + \bar{m}_0) \qquad \mu_0(m) < 0 \quad \textit{for all } m \in [0, \bar{m}_0]$$

Proof: Appendix A.3.

These riskless solution components are inputs to the next section's solution with risk.

4.2 Properties of the Model with Risk

We now return to the case with risk, $\sigma > 0$. When households face risk, the stochastic HJB (12) is second-order in the value function, so its envelope equation is second-order in marginal utility. To cleanly distinguish the risky model from the riskless model, we write relevant functions with σ subscripts, e.g., marginal utility is $\lambda_\sigma(m) \equiv v_\sigma'(m)$, the drift is $\mu_\sigma(m) \equiv \eta - \iota m - c_\sigma(m)$, etc. Lemma 3 gives the normalized envelope condition.

Lemma 3 *The normalized marginal utility λ_σ satisfies*

$$(\rho + \iota)\lambda_\sigma = \mu_\sigma(m)\lambda_\sigma' + \tfrac{\sigma^2}{2}\lambda_\sigma'' \tag{21}$$

Proof: Appendix A.10.

For $\sigma > 0$, the problem becomes more complicated because the envelope condition (21) is now a second-order ODE. In order to prove general results, our strategy is to approximate the model around the riskless case. Thus, our conclusions apply for low levels of risk.

This is a helpful approach because we know the riskless normalized consumption function $c_0(m)$ in closed form, and equations (81) and (82) in Appendix A.11 give the perturbation $\hat{c}$ in closed form. The following proposition describes household behavior under risk and states that this is an accurate approximation for small σ.

Proposition 5 *For $\sigma > 0$ sufficiently small and each $m \in [0, \bar{m}_0)$,*

$$c_\sigma(m) = c_0(m) + \sigma^2\hat{c}(m) + O(\sigma^4) \qquad c_\sigma'(m) = c_0'(m) + \sigma^2\hat{c}'(m) + O(\sigma^4)$$

The reset and upper boundaries satisfy

$$m_\sigma^* = m_0^* + O(\sigma^2) \qquad \bar{m}_\sigma = \bar{m}_0 + O(\sigma^2)$$

At the upper boundary, $c_\sigma(\bar{m}_\sigma) = \rho(1+\bar{m}_\sigma)$*. Consumption falls from approximately* $\bar{c}_0$ *to this value within an interval of width* $O(\sigma^2)$ *immediately below* $\bar{m}_\sigma$*. In particular,* $c_\sigma'(m) < 0$ *close to* $\bar{m}_\sigma$*.*

Proof: Appendix A.11.

Proposition 5 implies that the consumption function under low risk is very similar to the riskless consumption function with one exception. The riskless function $c_0(m)$ is monotonically increasing in m, but this is not true of $c_\sigma(m)$. Near the upper boundary, the function reverses as households are willing to increase savings in order to move up their likely illiquidation. Mathematically, this occurs because of the smooth pasting condition, which is absent in the riskless model where households never hit the upper boundary $\bar{m}_0$ from below.

Still, this discontinuity is limited to a small neighborhood around $\bar{m}_\sigma$. Everywhere else, including at and below the reset point m_σ^*, consumption behaves qualitatively the same as in the riskless model. We formalize this result in the main theorem:

Theorem 1 *The marginal propensity to consume (MPC) out of money has the following properties:*

1. *The MPC depends only on the ratio of money to illiquid wealth* $m = M/A$*.*
2. *For* σ *sufficiently small, the MPC exceeds the frictionless MPC* ρ *for all* $m \in [0, m_\sigma^*]$*.*
3. *For* σ *sufficiently small, the MPC is decreasing in* m *on* $[0, \bar{m}_\sigma]$*.*
4. *For* σ *sufficiently small, the MPC is negative in an* $O(\sigma^2)$*-wide neighborhood of* $\bar{m}_\sigma$*.*

Proof: Appendix A.12.

Proposition 6 (Risky iMPC) *For* $\sigma > 0$*, define expected future consumption per current asset by*

$$\mathcal{U}_\sigma(m,h) \equiv \mathbb{E}_m\left[\frac{C_h}{A_0}\right], \qquad \mathcal{I}_\sigma(m,h) \equiv \partial_m \mathcal{U}_\sigma(m,h).$$

Let $\mathcal{I}_0(m,h)$ *denote the corresponding object in the riskless model with the same income allocation* η*. Inside the inaction region,* $\mathcal{U}_\sigma$ *is characterized by*

$$\partial_h \mathcal{U}_\sigma = \alpha \mathcal{U}_\sigma + \mu_\sigma(m)\partial_m \mathcal{U}_\sigma + \frac{\sigma^2}{2}\partial_{mm}\mathcal{U}_\sigma, \qquad \mathcal{U}_\sigma(m,0) = c_\sigma(m), \tag{22}$$

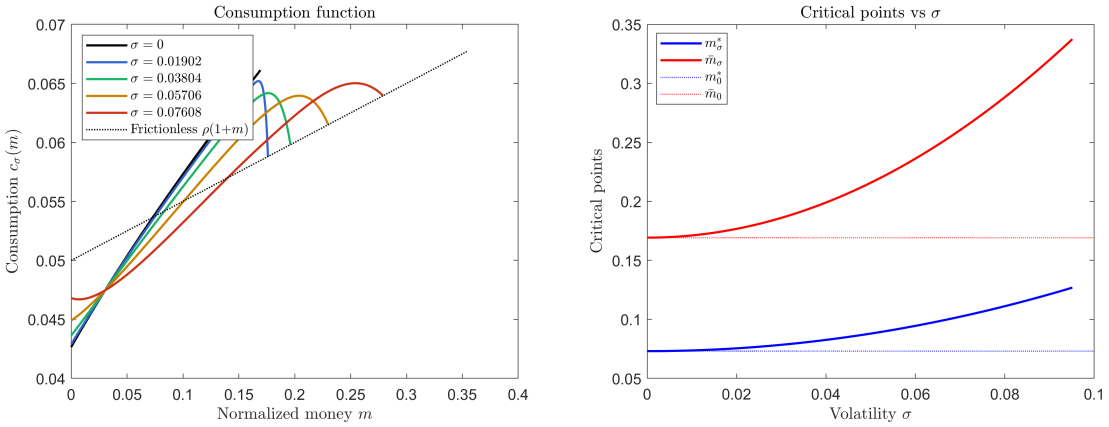


Figure 5: Consumption Function and Critical Points Under Income Risk

Note: Each panel shows the small-risk approximation in Appendix Lemma 10 for several σ values. *Left:* Consumption function $c_\sigma(m)$; the dotted curve is the frictionless benchmark $\rho(1+m)$; each curve terminates at $\bar{m}_\sigma$ where smooth pasting holds. *Right:* Reset threshold m^*_σ and upper boundary $\bar{m}_\sigma$ as functions of σ; dotted lines are the riskless benchmarks.

with reset boundary conditions, for $h > 0$,

$$\mathcal{U}_\sigma(0,h) = \frac{\mathcal{U}_\sigma(m^*_\sigma,h)}{1+m^*_\sigma}, \tag{23}$$

$$\mathcal{U}_\sigma(\bar{m}_\sigma,h) = \frac{1+\bar{m}_\sigma}{1+m^*_\sigma}\mathcal{U}_\sigma(m^*_\sigma,h). \tag{24}$$

The horizon-zero iMPC is the impact MPC:

$$\mathcal{I}_\sigma(m,0) = c'_\sigma(m).$$

For each $m < \bar{m}_0$ *and each horizon* h *that is not a riskless reset time,*

$$\lim_{\sigma\to 0}\mathcal{I}_\sigma(m,h) = \mathcal{I}_0(m,h)$$

Proof: Appendix A.13.

5 Aggregation

This section describes how household consumption and money demand affect the macroeconomy. First, we introduce the joint distribution of money and assets. Then we prove Theorem 2, which says that the MPC aggregates perfectly. Next, we describe the "Aggregating Measure", which gives a shortcut for characterizing aggregate dynamics from a one-dimensional KFE. Finally, we derive

equations describing aggregate dynamics.

5.1 The Dynamic Distribution

$G(M, A, t)$ denotes the joint density of money and illiquid assets in the economy at time t. The Kolmogorov forward equation (KFE) for this density is

$$\partial_t G = \partial_M^2 \left(\frac{\sigma^2 A^2}{2} G \right) - \partial_M ((\eta A - \pi M - C(M, A))G) - \partial_A(\alpha A G) + R(M, A, t) \tag{25}$$

where $R(M, A, t)$ is the flow of resetting agents to position M, A at time t. Given the optimizing behavior characterized thus far, this flow will be zero except at optimal reset portfolios.

The boundary conditions for this problem impose that the boundaries of the inaction region constitute absorbing barriers. These boundaries are the curves $M = 0$ and $M = \bar{M}(A)$, where $\bar{M}(A)$ gives the upper bound of the inaction region for a given asset level A. The relevant boundary conditions are that the density is zero at these barriers:

$$G(0, A, t) = 0 \tag{26}$$

$$G(\bar{M}(A), A, t) = 0 \tag{27}$$

There is also an initial condition

$$G(M, A, 0) = \Psi(M, A) \tag{28}$$

From G, we also define a joint distribution in (m, A) by $\tilde{G}(m, A, t) \equiv AG(mA, A, t)$.[6] Thus the unweighted marginal density $g(m, t)$ is

$$g(m, t) = \int_0^\infty \tilde{G}(m, A, t) dA \tag{29}$$

5.2 MPC Aggregation

We define the aggregate marginal propensity to consume $\mathbf{MPC}(t)$ as the marginal effect on aggregate consumption of giving every household the same additional dollar of real balances, holding prices and policy functions fixed. Before any portfolio adjustment, this shifts individual holdings from M to $M + \varepsilon$, so the transported density is $G(M - \varepsilon, A, t)$. Since the density vanishes at the adjustment boundaries, households who cross a boundary do not contribute to aggregate consumption at first

[6] The A coefficient on G is inherited from the change of variables. For intuition, observe that $1 = \int_0^\infty \int_0^\infty G(M, A, t) dA dM = \int_0^\infty \int_0^\infty AG(mA, A, t) dA dm$.

order. Thus the relevant first-order change in the interior density is $-G_M(M, A, t)$:

$$\mathbf{MPC}(t) \equiv -\int_0^\infty \int_0^{\bar{M}(A)} C(M, A) G_M(M, A, t) dM dA \tag{30}$$

Theorem 2 (Aggregation) *For $\sigma > 0$ and $t > 0$, the aggregate MPC from an equal-dollar transfer equals the number-weighted average individual MPC.*

Proof. Apply integration by parts to the interior integral of equation (30):

$$C(M, A)G(M, A, t)\Big|_0^{\bar{M}(A)} = \int_0^{\bar{M}(A)} C(M, A)G_M(M, A, t)\, dM + \int_0^{\bar{M}(A)} C_M(M, A)G(M, A, t)\, dM$$

The aggregate MPC becomes

$$\mathbf{MPC}(t) = \int_0^\infty \int_0^{\bar{M}(A)} C_M(M, A)G(M, A, t)\, dM\, dA - \int_0^\infty C(M, A)G(M, A, t)\Big|_0^{\bar{M}(A)} dA \tag{31}$$

The first term is the number-weighted average MPC. The boundary conditions (26) and (27) imply that $G(0, A, t) = G(\bar{M}(A), A, t) = 0$, so the second term is zero. ■

The weighting depends on the experiment. The theorem considers an equal-dollar transfer, so every household receives the same weight. A proportional transfer $dM = \varepsilon A$ instead shifts every household's normalized balance m by the same amount. The marginal consumption response to the shock is then the asset-weighted average MPC, calculated using the distribution in Section 5.3. Section 6 uses the same experiment to recover the full marginal response of aggregate money from steady-state adjustment statistics.

The fixed-cost structure is crucial for this exact aggregation result. Households can reset their portfolios as soon as they reach either boundary, so the usual boundary conditions imply that the density vanishes there. Appendix B shows that the same argument applies to a broader set of consumption and adjustment rules. By contrast, in models such as Kaplan and Violante (2014), portfolio adjustment opportunities arrive randomly, so households can accumulate at the zero-liquidity constraint. The boundary term in equation (31) then need not vanish, and the number-weighted average MPC need not equal the aggregate MPC.

5.3 The Aggregating Measure

The marginal density of m across households is not particularly useful for aggregation. For example, average m across households is the average liquidity ratio $\frac{M}{A}$ rather than any monetary aggregate. Weighting the density by assets is more promising, but an even more practical approach is to weight

by assets but abandon the normalization required to recover a probability measure. Doing so gives the *aggregating measure*, defined as

$$\mathcal{A}(m,t) = \int_0^\infty A\tilde{G}(m,A,t)dA \tag{32}$$

Thus the aggregating measure $\mathcal{A}$ weights the density by A.

The aggregating measure $\mathcal{A}$ is useful because it contains information about the macroeconomic aggregates. Aggregate illiquid assets, money, and consumption are easily written in terms of the aggregating measure by

$$\mathbf{A}(t) = \int_0^\infty \int_0^{\bar{m}} A\tilde{G}(m,A,t)dmdA = \int_0^{\bar{m}} \mathcal{A}(m,t)dm \tag{33}$$

$$\mathbf{M}(t) = \int_0^\infty \int_0^{\bar{m}} mA\tilde{G}(m,A,t)dmdA = \int_0^{\bar{m}} m\mathcal{A}(m,t)dm \tag{34}$$

$$\mathbf{C}(t) = \int_0^\infty \int_0^{\bar{m}} c(m)A\tilde{G}(m,A,t)dmdA = \int_0^{\bar{m}} c(m)\mathcal{A}(m,t)dm \tag{35}$$

and normalizing the aggregating measure gives the asset-weighted probability measure $\frac{\mathcal{A}(m,t)}{\mathbf{A}(t)}$.

5.4 Dynamics of the Aggregating Measure

This subsection describes how the aggregating measure evolves over time. Before giving the relevant KFE, some notation must be defined.

When a household with balance m resets its portfolio, its illiquid assets change from A to A'; the relationship is implied by the portfolio conservation equation (10):

$$(1+m)A = (1+m^*)A'$$

Define the reset ratios as

$$\psi_0 \equiv \frac{A'}{A}\bigg|_{m=0} = \frac{1}{1+m^*} \qquad \psi_{\bar{m}} \equiv \frac{A'}{A}\bigg|_{m=\bar{m}} = \frac{1+\bar{m}}{1+m^*} \tag{36}$$

The rate at which households reset their portfolios is dynamic. Let $F_0(A,t)$ and $F_{\bar{m}}(A,t)$ denote the "reset flows", i.e. the rate of adjustment through the lower and upper boundaries for households with pre-adjustment assets A:

$$F_0(A,t) \equiv \frac{\sigma^2}{2}\partial_m \tilde{G}(0,A,t) \qquad F_{\bar{m}}(A,t) \equiv -\frac{\sigma^2}{2}\partial_m \tilde{G}(\bar{m},A,t)$$

The asset-weighted boundary flows are

$$\begin{aligned}\mathbf{F}_0^A(t) &\equiv \int_0^\infty AF_0(A,t)\,dA = \frac{\sigma^2}{2}\partial_m \mathcal{A}(0,t)\\ \mathbf{F}_{\bar{m}}^A(t) &\equiv \int_0^\infty AF_{\bar{m}}(A,t)\,dA = -\frac{\sigma^2}{2}\partial_m \mathcal{A}(\bar{m},t)\end{aligned} \tag{37}$$

The next lemma gives a KFE that describes the dynamics of the aggregating measure.

Lemma 4 (Asset-Weighted KFE) *If $\sigma > 0$ and aggregate illiquid wealth is finite, $\mathbf{A}(t) < \infty$, the aggregating measure satisfies*

$$\begin{aligned}\partial_t \mathcal{A} = \frac{\sigma^2}{2}\partial_m^2 \mathcal{A} - \partial_m(\mu(m)\mathcal{A}) + \alpha\mathcal{A}\\ + \delta(m-m^*)\left[\psi_0 \mathbf{F}_0^A(t) + \psi_{\bar{m}}\mathbf{F}_{\bar{m}}^A(t)\right]\end{aligned} \tag{38}$$

with boundary conditions

$$\mathcal{A}(0,t) = \mathcal{A}(\bar{m},t) = 0$$

The initial condition is

$$\mathcal{A}(m,0) = \int_0^\infty A^2\Psi(mA,A)\,dA \tag{39}$$

Proof: Appendix A.14.

5.5 Aggregate Dynamics

Equations (33), (34), and (35) express aggregate assets, money, and consumption in terms of the aggregating measure. The measure is dynamic, and its KFE implies a concise dynamic equation for each variable.

Two reset-flow objects will be useful. The net aggregate liquidation flow of assets into money is

$$\mathbf{L}(t) \equiv (1-\psi_0)\mathbf{F}_0^A(t) + (1-\psi_{\bar{m}})\mathbf{F}_{\bar{m}}^A(t) \tag{40}$$

At a reset from boundary $b \in \{0, \bar{m}\}$, the change in money is $\Delta M = (1-\psi_b)A$. Thus $\mathbf{L}(t)$ is also the aggregate signed money flow generated by portfolio resets.

We will also use the gross aggregate flow of post-reset balances:

$$\mathbf{B}(t) \equiv m^*\left(\psi_0 \mathbf{F}_0^A(t) + \psi_{\bar{m}}\mathbf{F}_{\bar{m}}^A(t)\right) \tag{41}$$

This represents the balance-scaled flow, because a reset from boundary b leaves the household with assets $A' = \psi_b A$ and money $B = m^* A'$.

Consumption dynamics also depend on a precautionary term

$$\mathbf{P}(t) \equiv \int_0^{\bar{m}} \frac{c'(m)^2}{c(m)} \mathcal{A}(m,t)\, dm \tag{42}$$

Proposition 7 (Aggregate Laws of Motion) *If $\sigma > 0$, aggregate illiquid assets, money, and consumption satisfy*

$$\dot{\mathbf{A}}(t) = \alpha \mathbf{A}(t) - \mathbf{L}(t) \tag{43}$$

$$\dot{\mathbf{M}}(t) = \eta \mathbf{A}(t) - \pi \mathbf{M}(t) - \mathbf{C}(t) + \mathbf{L}(t) \tag{44}$$

$$\dot{\mathbf{C}}(t) = -(\rho + \pi)\mathbf{C}(t) + \sigma^2 \mathbf{P}(t) + (\rho - c(0))\, \mathbf{F}_0^A(t) \tag{45}$$

Proof: Appendix A.15.

Portfolio resets only reallocate wealth between money and illiquid assets, so $\mathbf{L}(t)$ enters the first two equations with opposite signs. The gross balance flow $\mathbf{B}(t)$ does not enter these laws because only the net money created at a reset changes aggregate stocks. It will instead be useful for describing the sequence of future resets below.

The consumption equation resembles an aggregate Euler equation for an asset with return $-\pi$, except with the addition of two terms. The precautionary term $\sigma^2 \mathbf{P}(t)$ appears because introducing risk transforms the Euler equation into a second-order differential equation. The lower reset flow $\mathbf{F}_0^A(t)$ (gross liquidations) also enters the consumption equation, but the upper reset flow does not because smooth-pasting makes consumption continuous at that boundary.

6 The Functional Sufficient Statistic

This section introduces the functional sufficient statistic. It is an expression for the impulse response of real balances to an aggregate monetary shock, given in terms of objects that can be calculated in the steady-state distribution. The derivation expands on the strategy developed in Adams (2026).

The sufficient statistic also applies beyond the household model developed above. Appendix B allows for an arbitrary consumption function, state-dependent risk, additional adjustment opportunities, and a non-zero lower adjustment threshold.

6.1 Definitions

Deriving the sufficient statistic requires defining new functions describing portfolio adjustment times. To do so, we also transform several model variables to map to objects that can be reasonably

measured in transaction data.

We study an aggregate shock that occurs at time $t = 0$. Index the reset that began the current inaction spell by $k = 0$ and subsequent resets by $k \geq 1$. Let B_k denote money immediately after reset k, and let a denote the age of the current spell. Define the "scaled initial balance" Z and relative money balance x by

$$Z \equiv B_0 e^{\alpha a} \qquad x \equiv \frac{M(0)}{Z} \tag{46}$$

$A(-a)$ denotes the household's illiquid assets immediately after reset zero. Thus $B_0 = m^* A(-a)$, and because assets grow at rate α, $A(0) = A(-a)e^{\alpha a}$. Together, these imply $Z = m^* A(0)$ and $x = m(0)/m^*$. Weighting by Z is therefore equivalent to weighting by assets, and every adjustment resets to $x = 1$. Let $f_Z(Z)$ denote the initial marginal density of spell scales. The aggregate spell scale at $t = 0$ is

$$\mathbf{Z}_0 \equiv \int_0^\infty Z f_Z(Z)\, dZ = m^* \mathbf{A}_0 \tag{47}$$

Z and x are useful transformations of model variables because they can be constructed from observed money balances and reset times. We discuss in Section 7 how to calculate α.

The monetary shock is a transfer $\varepsilon_m A$ that gives each household money in proportion to their assets. Thus all households become equally more liquid, increasing their normalized money m by ε_m. In terms of the transformed variables, the shock increases relative money x by $\varepsilon_x \equiv \frac{\varepsilon_m}{m^*}$. Therefore, for any aggregate outcome $\mathbf{Y}$,

$$\frac{1}{\mathbf{Z}_0}\frac{\partial \mathbf{Y}}{\partial \varepsilon_x} = \frac{1}{\mathbf{A}_0}\frac{\partial \mathbf{Y}}{\partial \varepsilon_m}$$

The two representations give the same marginal response to the shock.

Define the adjustment

$$\Delta M \equiv B_k - M^- \tag{48}$$

where M^- denotes money immediately before a reset, and the reset balance B_k measures money immediately afterward. A liquidation has $\Delta M > 0$, while an illiquidation has $\Delta M < 0$.

Let τ denote the time until the next reset. To describe its timing and size, define the cumulative functions

$$\begin{aligned} \mathcal{F}_B(t \mid x) &\equiv \mathbb{E}\left[\frac{B_1}{Z}\mathbf{1}\{\tau \leq t\} \,\middle|\, x\right] \\ \mathcal{F}_J(t \mid x) &\equiv \mathbb{E}\left[\frac{\Delta M}{Z}\mathbf{1}\{\tau \leq t\} \,\middle|\, x\right] \end{aligned} \tag{49}$$

$\mathcal{F}_B$ is the steady-state expected post-reset balance, relative to initial scale, generated by the next reset when it occurs by t. $\mathcal{F}_J$ is the corresponding expected adjustment.

Define their time derivatives

$$f_B(t \mid x) \equiv \partial_t \mathcal{F}_B(t \mid x)$$
$$f_J(t \mid x) \equiv \partial_t \mathcal{F}_J(t \mid x) \tag{50}$$

An adjustment resets to $x = 1$, so define the functions at $x = 1$ as

$$f_B^*(t) \equiv f_B(t \mid 1) \qquad f_J^*(t) \equiv f_J(t \mid 1) \tag{51}$$

The marginal effects of the ε_x transfer on the average conditional adjustment functions are

$$D_B(t) \equiv \int \partial_x f_B(t \mid x) \bar{h}_Z(x)\, dx$$
$$D_J(t) \equiv \int \partial_x f_J(t \mid x) \bar{h}_Z(x)\, dx \tag{52}$$

where $\bar{h}_Z(x)$ denotes the stationary density of x when households are weighted by Z.

Let $\mathbf{B}(t; \varepsilon_x)$ and $\mathbf{L}(t; \varepsilon_x)$ denote the aggregate post-reset balance and net liquidation flows after the $\varepsilon_x Z$ transfer. Their marginal responses are

$$\mathbf{B}^{\varepsilon_x}(t) \equiv \frac{1}{\mathbf{Z}_0} \left. \frac{\partial \mathbf{B}(t; \varepsilon_x)}{\partial \varepsilon_x} \right|_{\varepsilon_x = 0}$$
$$\mathbf{L}^{\varepsilon_x}(t) \equiv \frac{1}{\mathbf{Z}_0} \left. \frac{\partial \mathbf{L}(t; \varepsilon_x)}{\partial \varepsilon_x} \right|_{\varepsilon_x = 0} \tag{53}$$

Equations (40) and (41) give their model representations. Empirically, $\mathbf{B}(t)$ is the sum of post-reset balances per unit of time, while $\mathbf{L}(t)$ is the corresponding sum of signed money adjustments. Neither construction requires the reset parameters.

Lemma 5 (Reset-flow response) *The post-reset balance flow and the net liquidation flow satisfy*

$$\mathbf{B}^{\varepsilon_x}(t) = D_B(t) + \int_0^t f_B^*(t-s) \mathbf{B}^{\varepsilon_x}(s)\, ds \tag{54}$$
$$\mathbf{L}^{\varepsilon_x}(t) = D_J(t) + \int_0^t f_J^*(t-s) \mathbf{B}^{\varepsilon_x}(s)\, ds \tag{55}$$

Proof: Appendix A.16.

6.2 The Marginal Effect on Aggregates

Aggregating the monetary shock requires two functional objects. The first is the net liquidation response $\mathbf{L}^{\varepsilon_x}(t)$ determined by Lemma 5. Because $\varepsilon_x = \varepsilon_m/m^*$ and $\mathbf{Z}_0 = m^*\mathbf{A}_0$, its marginal response to the transfer is

$$\mathbf{L}^{\varepsilon_m}(t) \equiv \frac{1}{\mathbf{A}_0} \left.\frac{\partial \mathbf{L}(t;\varepsilon_m)}{\partial \varepsilon_m}\right|_{\varepsilon_m=0} = \mathbf{L}^{\varepsilon_x}(t) \tag{56}$$

The second object is the average iMPC. Let $\text{iMPC}(m,t)$ denote the response of consumption at t to an additional dollar of money at date zero for a household beginning with normalized money m. This response incorporates any portfolio resets before t. Let $\bar{\mathcal{A}}(m)$ denote the aggregating measure in the initial steady state and define its asset-weighted marginal density by

$$\begin{aligned} \bar{g}_A(m) &\equiv \frac{\bar{\mathcal{A}}(m)}{\mathbf{A}_0} \\ \overline{\text{iMPC}}(t) &\equiv \int_0^{\bar{m}} \text{iMPC}(m,t)\bar{g}_A(m)\,dm \end{aligned} \tag{57}$$

Define the marginal responses of aggregate consumption, money, and illiquid assets per dollar of aggregate transfers by

$$\begin{aligned} \mathbf{C}^{\varepsilon_m}(t) &\equiv \frac{1}{\mathbf{A}_0} \left.\frac{\partial \mathbf{C}(t;\varepsilon_m)}{\partial \varepsilon_m}\right|_{\varepsilon_m=0} \\ \mathbf{M}^{\varepsilon_m}(t) &\equiv \frac{1}{\mathbf{A}_0} \left.\frac{\partial \mathbf{M}(t;\varepsilon_m)}{\partial \varepsilon_m}\right|_{\varepsilon_m=0} \\ \mathbf{A}^{\varepsilon_m}(t) &\equiv \frac{1}{\mathbf{A}_0} \left.\frac{\partial \mathbf{A}(t;\varepsilon_m)}{\partial \varepsilon_m}\right|_{\varepsilon_m=0} \end{aligned} \tag{58}$$

Theorem 3 (Functional sufficient statistic) *The aggregate consumption response satisfies*

$$\mathbf{C}^{\varepsilon_m}(t) = \overline{\text{iMPC}}(t) \tag{59}$$

The marginal responses of aggregate money and illiquid assets satisfy

$$\begin{aligned} \dot{\mathbf{M}}^{\varepsilon_m}(t) &= \eta \mathbf{A}^{\varepsilon_m}(t) - \pi \mathbf{M}^{\varepsilon_m}(t) - \overline{\text{iMPC}}(t) + \mathbf{L}^{\varepsilon_m}(t) \\ \dot{\mathbf{A}}^{\varepsilon_m}(t) &= \alpha \mathbf{A}^{\varepsilon_m}(t) - \mathbf{L}^{\varepsilon_m}(t) \end{aligned} \qquad \mathbf{M}^{\varepsilon_m}(0) = 1 \qquad \mathbf{A}^{\varepsilon_m}(0) = 0 \tag{60}$$

Lemma 5 determines $\mathbf{L}^{\varepsilon_m}$ *through equation* (56)*. In the benchmark case with* $\eta = 0$*,*

$$\mathbf{M}^{\varepsilon_m}(t) = e^{-\pi t} - \int_0^t e^{-\pi(t-s)} \overline{\text{iMPC}}(s)\,ds + \int_0^t e^{-\pi(t-s)} \mathbf{L}^{\varepsilon_m}(s)\,ds \tag{61}$$

Proof. A household with initial assets A and normalized money m receives $\varepsilon_m A$. By the definition of the iMPC,

$$\left.\frac{\partial \mathbb{E}[C(t;\varepsilon_m) \mid m, A]}{\partial \varepsilon_m}\right|_{\varepsilon_m=0} = A\,\mathrm{iMPC}(m,t) \tag{62}$$

Integrating over households and using the definition of the aggregating measure gives

$$\mathbf{C}^{\varepsilon_m}(t) = \frac{1}{\mathbf{A}_0}\int_0^{\bar{m}} \mathrm{iMPC}(m,t)\bar{\mathcal{A}}(m)\,dm = \overline{\mathrm{iMPC}}(t) \tag{63}$$

Differentiate equations (43)–(44) with respect to ε_m and divide by $\mathbf{A}_0$. Equation (56) and Lemma 5 give the marginal net liquidation flow. The transfer adds $\varepsilon_m \mathbf{A}_0$ to aggregate money and leaves illiquid assets unchanged. Since $\bar{g}_A(0) = \bar{g}_A(\bar{m}) = 0$, any resets on impact are second order or higher in ε_m. Thus $\mathbf{M}^{\varepsilon_m}(0) = 1$ and $\mathbf{A}^{\varepsilon_m}(0) = 0$. This proves equation (60). Solving its first equation when $\eta = 0$ gives equation (61). ■

The empirical inputs are the asset-weighted average iMPC, the two next-reset money functions $f_B(t \mid x)$ and $f_J(t \mid x)$, the stationary Z-weighted density of x, and the aggregate coefficients. Equations (51) and (52) determine the remaining terms. The two money functions incorporate the reset rule.

The theorem applies to a transfer proportional to A, which shifts every household's normalized money by the same amount. An equal-dollar transfer instead shifts m by an amount that depends on A, or x by an amount that depends on Z, and is not covered by the theorem.

7 Transaction-Level Evidence

This section describes our checking account data, reports descriptive statistics of balances and adjustment behavior, and estimates the functional sufficient statistic for the effects of a monetary shock.

7.1 Data

Our anonymous microdata come from several of the core account-processing platforms offered by the data provider. The main dataset used in our analysis comes from one such platform and includes transaction-level data for all 25 million deposit accounts associated with a set of 164 banks and credit unions. For each account, we observe all transactions into and out of the account, as well as details on each transaction. One can think of this loosely as observing the monthly bank statements for each account. We also observe details about the account owner, such as age, sex, and location, and can proxy for characteristics such as income because we observe direct deposits into each account.

Our panel covers the period 2022–2026.

We are primarily interested in liquidation and illiquidation events. We classify each transaction in our data as a liquidation event, an illiquidation event, or other. We classify as liquidation events any incoming transfers from a brokerage account or another bank account owned by the individual, as well as pension income. Conversely, we classify as illiquidation events any outgoing transfers to a brokerage account or another bank account owned by the individual. To account for the fact that some accounts are less active than others, we restrict our attention to accounts with at least two transactions per month. To limit the influence of outliers, we winsorize account balances and transfer amounts at the 1% level.

7.2 Summary Statistics and Adjustment Moments

Similar to Baley and Blanco (2021), Alvarez et al. (2024), or Baley and Blanco (2026), we summarize portfolio adjustment using the size and timing of liquidations and illiquidations. Index a household's adjustments by j. Let M_j^- and M_j^+ denote money immediately before and after adjustment j, and define the signed change in money by

$$\Delta M_j \equiv M_j^+ - M_j^-$$

A liquidation has $\Delta M_j > 0$, while an illiquidation has $\Delta M_j < 0$. Let τ_j denote the time since the household's previous adjustment. Table 1 reports our estimates.

Table 1: Money and Adjustment Statistics

Description	Statistic	Data
Money balances		
Average balance	$\mathbb{E}[M]$	\$ 14,715
Squared coefficient of variation	$\text{CV}^2[M]$	7.61
Frequency and duration		
Adjustment frequency	F	8.01
Liquidation frequency	F_L	4.60
Illiquidation frequency	F_I	3.41
Average duration (years)	$\mathbb{E}[\tau]$	0.125
Squared coefficient of variation	$\text{CV}^2[\tau]$	—
Adjustment size		
Average adjustment	$\mathbb{E}[\Delta M]$	\$823
Average liquidation	$\mathbb{E}[\Delta M \mid \Delta M > 0]$	\$ 3,005
Average illiquidation	$\mathbb{E}[\Delta M \mid \Delta M < 0]$	\$ -2,529
Second Moment	$\mathbb{E}[(\Delta M)^2]$	\$30,209,474
Skewness	$\text{Skew}[\Delta M]$	0.12
Kurtosis	$\text{Kur}[\Delta M]$	17.29

Note: F, F_L, and F_I denote adjustments, liquidations, and illiquidations per household per year. Money balances and adjustment sizes are measured in dollars.

7.3 The Empirical Functional Sufficient Statistic

Per Section 6, in order to calculate the functional sufficient statistic, we need to measure the cumulative reset functions (49)

$$\mathcal{F}_B(t \mid x) \equiv \mathbb{E}\left[\frac{B_1}{Z}\mathbf{1}\{\tau \leq t\} \,\middle|\, x\right]$$
$$\mathcal{F}_J(t \mid x) \equiv \mathbb{E}\left[\frac{\Delta M}{Z}\mathbf{1}\{\tau \leq t\} \,\middle|\, x\right]$$

where τ is the time until the next adjustment, $Z = B_0 e^{\alpha a}$ is the "scaled initial balance", and $x = \frac{M}{Z}$ is the "relative balance". M denotes a household's money balance, a is time since the last adjustment, and B_0 is their money balance immediately following the last adjustment.

This section describes how we estimate these functions.

7.3.1 Estimating α

Before we can estimate the conditional reset functions, we must construct the transformed balances Z and x, which requires choosing a value for the growth rate α. This parameter is straightforward to measure in the illustrative model, but it is more useful to have an approach that is consistent with the more general model in Appendix B to which the functional sufficient statistic applies.

Fortunately, we can estimate α from individuals' adjustment histories, accounting for liquidation and illiquidation events. The following proposition shows how α is related to observable objects in the generalized model. B_{k-1} and B_k denote money balances immediately after consecutive adjustments, τ_k the time between them, and ΔM_k the change in money at adjustment k.

Proposition 8 *In the generalized model, consecutive reset balances satisfy*

$$B_k = B_{k-1} e^{\alpha \tau_k} - m^* \Delta M_k \tag{64}$$

Proof: Appendix A.17.

Proposition 8 provides a useful estimation equation. For account i, dividing by the previous reset balance gives

$$\frac{B_{ik}}{B_{i,k-1}} = e^{\alpha \tau_{ik}} - m^* \frac{\Delta M_{ik}}{B_{i,k-1}} + u_{ik} \tag{65}$$

with residual u_{ik}. In practice, we assume that u_{ik} is normally distributed, and independent of observables τ_{ik} and $\frac{\Delta M_{ik}}{B_{i,k-1}}$. Then we estimate equation (65) by maximum likelihood.

Given this estimate, we can construct Z and x and proceed with estimating the cumulative reset functions.

7.3.2 Estimating the Cumulative Reset Functions

We care about the slopes of the cumulative reset functions, so we use a functional form that is linear in relative balance x, with exponential terms in the forward time horizon t. We estimate its parameters by Gaussian quasi-maximum likelihood. Index accounts by i and observation dates by s. Let τ_{is} be the time from date s to account i's next adjustment, $B_{1,is}$ represents balance immediately after that adjustment, and ΔM_{is} its signed size (ie. a liquidation has $\Delta M_{is} > 0$ and an illiquidation has $\Delta M_{is} < 0$.) The estimation equations are

$$\frac{B_{1,is}}{Z_{is}}\mathbf{1}\{\tau_{is} \le t\} = \gamma_B\left(1 - e^{-\lambda_B t}\right)x_{is} + \chi_B\left(1 - e^{-\eta_B t}\right) + u_{B,is}(t) \tag{66}$$

$$\frac{\Delta M_{is}}{Z_{is}}\mathbf{1}\{\tau_{is} \le t\} = \gamma_J\left(1 - e^{-\lambda_J t}\right)x_{is} + \chi_J\left(1 - e^{-\eta_J t}\right) + u_{J,is}(t) \tag{67}$$

where the errors have finite second moments and satisfy

$$\mathbb{E}\left[u_{K,is}(t) \mid x_{is}\right] = 0 \qquad K \in \{B, J\}$$

Taking conditional expectations (i.e. omitting the $u(t)$ errors) therefore recovers $\mathcal{F}_B(t \mid x)$ and $\mathcal{F}_J(t \mid x)$. Both left-hand sides equal zero if no adjustment occurs by t. Once the next adjustment occurs, each outcome records that adjustment alone, even if further adjustments occur before t.

$$(\gamma_B, \lambda_B, \chi_B, \eta_B) = (0.15, 16, 0.8, 16)$$
$$(\gamma_J, \lambda_J, \chi_J, \eta_J) = (-0.45, 16, 0.61, 16)$$

7.3.3 Estimated Functional Sufficient Statistic

Theorem 2 says that MPCs aggregate perfectly: the aggregate consumption response is equivalent to the average iMPC $\overline{\text{iMPC}}(t)$. Therefore, we lean on the large empirical literature that estimates the iMPC curve. For our main specification, we rely on the estimates from Fagereng et al. (2021), which measure the consumption effects of lottery windfalls. They estimate an MPC of 0.52 in the lottery-winning year, approximately 0.20 in the following year, and approximately 0.90 cumulatively over five years. To fit this pattern, we assume

$$\overline{\text{iMPC}}(t) = \text{MPC}_0\, e^{-\lambda_C t}$$

This yields $\text{MPC}_0 = 0.749$ and $\lambda_C = 0.814$ when t is measured in years.

Figure 6 plots the marginal responses to a transfer that gives households money in proportion to their illiquid assets. Consumption is the assumed average iMPC implied by the Fagereng et al. (2021) estimates. The liquidation flow follows from the estimated reset functions, and the illiquid asset and money responses follow from Theorem 3.

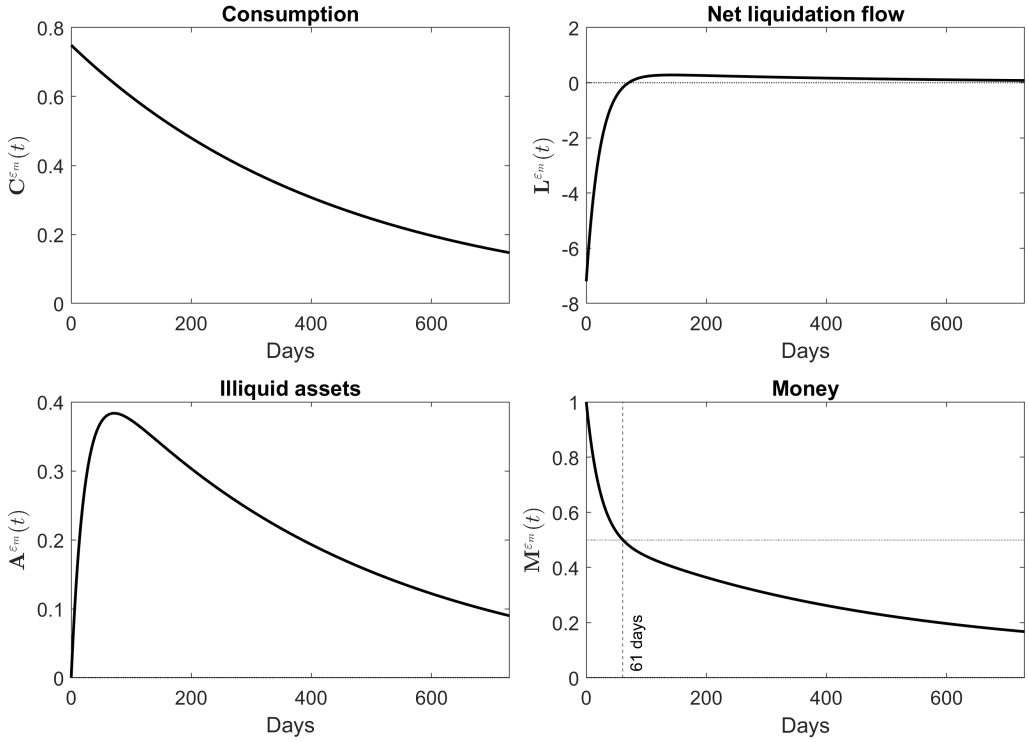


Figure 6: The Functional Sufficient Statistic

Note: Horizons are in days. Consumption and net liquidation are flow responses per dollar transferred per year. Illiquid assets and money are stock responses per dollar transferred.

Our estimated functional sufficient statistic implies that money demand is quantitatively sluggish. Figure 6 demonstrates this clearly in the fourth panel, where the half-life of real balances to a monetary shock is roughly 60 days. This result strongly rejects static models such as money-in-utility; monetary expansions take a long time to return to steady state because households adjust balances relatively slowly.

7.4 Model Calibration

We use the observed adjustment behavior from our transaction-level data to calibrate the model. First, we set inflation $\pi = .04$ to match the annual CPI inflation over our 2022-2024 sample period. We set $i = .09$ to match the average marginal nominal rate in the 2022 Survey of Consumer Finances

(SCF).[7] We set the discount rate to match the implied real rate $\rho = r = .05$ so that our economy converges to a steady state. This real rate is larger than conventional calibrations, but is consistent with the marginal safe real return facing average households. Moreover, the discount rate represents the MPC out of illiquid income in our model, and $\rho = .05$ is consistent with the typical estimates which tend to range from 3% - 7% (Mian et al., 2013; Aladangady, 2017; Di Maggio et al., 2020; Chodorow-Reich et al., 2021). We choose $(\sigma, \eta, \varkappa) = (0.0571, 0.0089, 0.1084)$ to match an aggregate money-to-illiquid-asset ratio of 6.2%, an adjustment-size ratio of 1.331, and a money-response half-life of approximately 60 days. The size ratio is average dollar illiquidation divided by average dollar liquidation. Table 2 reports the parameters and implied outcomes. Appendix G reports their local sensitivity to each model parameter.

Table 2: Baseline Calibration

Description	Parameter	Value
Primitive parameters		
Discount rate	ρ	0.050
Real return	r	0.050
Inflation rate	π	0.040
Liquid income rate	η	0.009
Brownian std. dev.	σ	0.057
Utility adjustment cost	$\varkappa$	0.10844
Derived parameters		
Nominal interest rate	$i = \pi + r$	0.090
Illiquid income share	$\alpha = r - \eta$	0.041
Composite rate	$\iota = i - \eta$	0.081
Implied model outcomes		
Optimal reset ratio	m^*	0.073
Upper boundary ratio	$\bar{m}$	0.169
Liquidation frequency	F_L	0.686
Illiquidation frequency	F_I	0.023
Average duration	$\mathbb{E}[\tau]$ (years)	1.411
Money-to-asset ratio	$\mathbf{M}/\mathbf{A}$	0.0620
Adjustment-size ratio (I/L)	R_Δ	1.331
Money-response half-life	$h_{1/2}$ (days)	59.844
Average MPC out of money	$\overline{MPC}$	0.151

Note: The parameter $\varkappa$ is the fixed utility cost of adjustment. Rates are annualized and the half-life is reported in days. Reset ratios and the average MPC are computed from the closed-form expressions in Section 3. Adjustment frequencies and average duration use the linear KFE approximation.

[7] While Treasuries may pay 4%, this is not the marginal risk-free for most US households. It may be for the minority that are debt-free, but most US households can earn a higher risk-free return by paying down debt, even after accounting for maturity differences. As an example, for the 37% of US households in 2022 that rolled over credit card debt month-by-month, their marginal safe nominal rate was close to 20%. We calculate the marginal (highest) nominal rate, adjusted for maturity in the SCF, and estimate that the average across households in 2022 was 9.7%. Moreover, this is probably a conservative value; using credit bureau data, Fulford and Schuh (2024) estimate that the majority of Americans roll over credit card debt, substantially raising the average MPC.

8 Conclusion

In this paper, we studied a tractable theory of consumption-based money demand with endogenous consumption. Because households must use money to consume, money is their marginal asset between portfolio adjustments. This simple mechanism generates realistic MPCs that are large throughout the wealth distribution, higher for households with low liquidity, and slowly declining in the consumption horizon. Most surprisingly, the cashless limit is not cashless; as the adjustment cost becomes small and velocity rises, the MPC distortion does not disappear (unless the Friedman rule holds).

The same adjustment friction makes aggregate money demand state-dependent and sluggish. We derive a functional sufficient statistic that maps steady-state adjustment behavior and the average iMPC into the marginal response of aggregate money demand to a monetary shock. These objects can be constructed from transaction-level data without solving for the entire joint distribution of household portfolios. Then we estimate them using US transaction-level data.

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A Proofs

A.1 Proof of Proposition 1

Proof.

Conjecture separability:

$$V(mA, A) = v(m) + w(A)$$

which implies

$$v'(m) = V_M A \qquad w'(A) = V_M m + V_A \tag{68}$$
$$v''(m) = V_{MM} A^2$$

With this property, the HJB becomes (noting $\dot{m} = \eta - \alpha m - \pi m - c$, and $\iota = \alpha + \pi$)

$$\rho(v(m) + w(A)) = \max_c \log c + \log A + (\eta - \pi m - c)v'(m) + \alpha A(w'(A) - v'(m)m/A) + v''(m)\frac{\sigma^2}{2}$$

Or separately:

$$\rho v(m) = \max_c \log c + (\eta - \iota m - c)v'(m) + \frac{\sigma^2}{2}v''(m) \qquad \rho w(A) = \log A + \alpha A w'(A)$$

First solve for the asset component, which is a simple ODE with general solution

$$\rho w(A) = \log A + \frac{\alpha}{\rho} + B_A A^{\frac{\rho}{\alpha}}$$

$$\implies \rho w'(A) = \frac{1}{A} + \frac{\rho}{\alpha} B_A A^{\frac{\rho}{\alpha}-1}$$

The constant B_A is pinned by the transversality condition. Inside the inaction region, $A(t) = A_0 e^{\alpha t}$, so the homogeneous term satisfies

$$e^{-\rho t} B_A A(t)^{\rho/\alpha} = e^{-\rho t} B_A (A_0 e^{\alpha t})^{\rho/\alpha} = B_A A_0^{\rho/\alpha}$$

which is constant in t and does not vanish as $t \to \infty$ unless $B_A = 0$. Transversality requires $\lim_{t\to\infty} e^{-\rho t} V(M(t), A(t)) = 0$, so $B_A = 0$.

To find the stochastic process for m, normalize the stochastic processes for M (equation (3)) and for A (equation (2)) and apply the assumptions $y = 0$, and $s(A) = \sigma A$:

$$\frac{dM}{A} = \left(\eta - \pi\frac{M}{A} - \frac{C}{A}\right)dt - \sigma dW^q$$

$$\frac{dA}{A} = \alpha dt$$

Then use the definitions for m and c to get

$$dm = \frac{dM}{A} - \frac{M}{A}\frac{dA}{A} = (\eta - (\alpha + \pi)m - c)dt - \sigma dW^q$$

To find the optimal reset condition, use the derivatives in (68) to write

$$V_M = V_A \implies v'(m)\frac{1}{A} = w'(A) - v'(m)\frac{m}{A}$$

which for $\rho w(A) = \log A + \frac{\alpha}{\rho} + B_A A^{\frac{\rho}{\alpha}}$ is

$$v'(m)(1+m) = \frac{1}{\rho} + \frac{B_A}{\alpha}A^{\frac{\rho}{\alpha}}$$

The first-order condition for c is

$$\frac{1}{c} = v'(m)$$

Combined with transversality ($B_A = 0$) this implies

$$v'(m)(1+m) = \frac{1}{\rho}$$

$$\implies c^* = \rho(1+m^*)$$

And value matching is

$$v(0) + \frac{1}{\rho}\log(A) = v(m^*) + \frac{1}{\rho}\log(A^*) - \varkappa$$

Use the portfolio constraint $A = A^* + M^*$

$$v(0) + \frac{1}{\rho}\log(A^* + m^*A^*) = v(m^*) + \frac{1}{\rho}\log(A^*) - \varkappa$$

$$v(0) + \frac{1}{\rho}\log(1+m^*) = v(m^*) - \varkappa$$

The other value matching condition $V(\bar{M}, A) = V(M^*, A^*) - \varkappa$ is

$$v(\bar{m}) + \frac{1}{\rho}\log(A) = v(m^*) + \frac{1}{\rho}\log(A^*) - \varkappa$$

Use the portfolio constraint $(1+\bar{m})A = A^* + M^*$

$$v(\bar{m}) + \frac{1}{\rho}\log\left(\frac{A^* + m^*A^*}{1+\bar{m}}\right) = v(m^*) + \frac{1}{\rho}\log(A^*) - \varkappa$$

$$v(\bar{m}) + \frac{1}{\rho}\log(1+m^*) - \frac{1}{\rho}\log(1+\bar{m}) = v(m^*) - \varkappa$$

For smooth-pasting (18), apply (68) to the unnormalized condition (11): $V_M = v'(\bar{m})/A$ and $V_A = w'(A) - v'(\bar{m})\bar{m}/A$, so $V_M = V_A$ gives

$$v'(\bar{m})\frac{1+\bar{m}}{A} = w'(A)$$

$B_A = 0$ implies $w'(A) = 1/(\rho A)$, thus

$$v'(\bar{m}) = 1/(\rho(1+\bar{m}))$$

and $c = 1/v'$ gives equation (18).

■

A.2 Proof of Lemma 1

Proof.

The MPC is $C_M(M, A)$, the partial derivative of the consumption function with respect to money. To find this derivative, the definition $c = \frac{C}{A}$ implies that the consumption function is related to the normalized $c(m)$ function by $C(M, A) = Ac(m)$. The $c'(m)$ derivative is equivalent to the MPC:

$$c'(m) = \frac{d}{dm}\left(\frac{1}{A}C(mA, A)\right) = C_M(M, A) \tag{69}$$

■

A.3 Proof of Lemma 2

We first derive the riskless consumption function for arbitrary η.

Lemma 6 *In the riskless model, for $\iota \neq 0$ and $\rho + \iota \neq 0$, consumption and money evolve by*

$$c(t) = c^* e^{-(\rho+\iota)t} \qquad m(t) = \frac{c(t)}{\rho} - \frac{i}{\iota}\left(\frac{c(t)}{c^*}\right)^{\frac{\iota}{\rho+\iota}} + \frac{\eta}{\iota}$$

where t is time since the last reset. The level and derivative of the inverse consumption function $m(c)$ wherever it exists are

$$m(c) = \frac{c}{\rho} - \frac{i}{\iota}\left(\frac{c}{c^*}\right)^{\frac{\iota}{\rho+\iota}} + \frac{\eta}{\iota} \qquad m'(c) = \frac{1}{\rho} - \frac{i}{\rho+\iota}\frac{c^{\frac{-\rho}{\rho+\iota}}}{(c^*)^{\frac{\iota}{\rho+\iota}}} \tag{70}$$

If the path reaches the zero-balance boundary at time T, then T is related to m^* by

$$m^* = \frac{i}{\iota}e^{\rho T} - \frac{\eta}{\iota}e^{(\rho+\iota)T} - 1$$

Lemma 7 *Suppose $\sigma = 0$, $\iota \neq 0$, and $\rho + \iota \neq 0$. The optimal reset values satisfy*

$$c^* = \rho(1+m^*)$$

and m^* is characterized by

$$\frac{1}{\rho}\log(1+m^*) + \varkappa = \int_0^{m^*}\frac{1}{c(m)}dm \tag{71}$$

where $m(c)$ is given by Lemma 6 and $c(m)$ denotes its inverse.

Lemma 8 *In the riskless model with $\iota \neq 0$ and $\rho+\iota \neq 0$, if the riskless model has an upper boundary $\bar{m} > m^*$, it satisfies the value-matching condition*

$$\frac{1}{\rho}\log\frac{\bar{c}}{c^*} + \frac{i}{\rho}\left(\frac{\bar{c}^{-\rho/(\rho+\iota)}}{(c^*)^{\iota/(\rho+\iota)}} - \frac{1}{c^*}\right) = \frac{1}{\rho}\log\frac{1+\bar{m}}{1+m^*} - \varkappa \tag{72}$$

where $\bar{c} \equiv \lim_{m\uparrow\bar{m}} c(m)$ is determined by the inverse consumption function (70)*.*

Proof of Lemma 6.

Setting $\sigma = 0$ in the normalized HJB (12) and differentiating with respect to m gives

$$\rho v'(m) = -\iota v'(m) + (\eta - \iota m - c)v''(m)$$

Letting $\lambda(t) = v'(m(t))$ and using $\dot{m} = \eta - (\iota m + c)$, we have

$$\dot{\lambda} = v''(m)\dot{m} = v''(m)(\eta - \iota m - c)$$

so the envelope condition can be written in terms of marginal utility as

$$(\rho + \iota)\lambda = \dot{\lambda}$$

which converts to the consumption Euler equation through the first-order condition $\lambda = \frac{1}{c}$ and $\dot{\lambda} = -\frac{\dot{c}}{c^2}$:

$$\dot{c} = -(\rho + \iota)c$$

The solution to this ODE is

$$c(t) = c^* e^{-(\rho+\iota)t}$$

The law of motion for m is

$$\dot{m} = \eta - \iota m(t) - c(t)$$

With exponential $c(t)$, this becomes

$$\dot{m} + \iota m = \eta - c^* e^{-(\rho+\iota)t}$$

Using the integrating factor $e^{\iota t}$ gives the solution

$$m(t) = \frac{c(t)}{\rho} + \frac{\eta}{\iota} + B_c(c^*)^{\frac{\iota}{\rho+\iota}} e^{-\iota t}$$

for some B_c. The boundary condition is

$$m(0) = m^* = \frac{c^*}{\rho} + \frac{\eta}{\iota} + B_c(c^*)^{\frac{\iota}{\rho+\iota}} \tag{73}$$

$$\implies B_c = (m^* - \frac{c^*}{\rho} - \frac{\eta}{\iota})(c^*)^{-\frac{\iota}{\rho+\iota}} = -\left(1 + \frac{\eta}{\iota}\right)(c^*)^{-\frac{\iota}{\rho+\iota}}$$

where the final equality follows from the optimal reset $c^* = \rho(1 + m^*)$. We can use $i = \pi + r = \iota + \eta$ to simplify further as

$$B_c = -\frac{i}{\iota}(c^*)^{-\frac{\iota}{\rho+\iota}}$$

Then $m(t)$ can be written as

$$m(t) = \frac{c(t)}{\rho} - \frac{i}{\iota}e^{-\iota t} + \frac{\eta}{\iota} = \frac{c(t)}{\rho} - \frac{i}{\iota}\left(\frac{c(t)}{c^*}\right)^{\frac{\iota}{\rho+\iota}} + \frac{\eta}{\iota}$$

This can equivalently be defined as an inverse consumption function on the branch $c \in [c(T), c^*]$:

$$m(c) = \frac{c}{\rho} - \frac{i}{\iota}\left(\frac{c}{c^*}\right)^{\frac{\iota}{\rho+\iota}} + \frac{\eta}{\iota} \tag{74}$$

Taking the derivative gives the desired expression.

At time T, money balances run out:

$$m(T) = 0 = \frac{c(T)}{\rho} - \frac{i}{\iota}\left(\frac{c(T)}{c^*}\right)^{\frac{\iota}{\rho+\iota}} + \frac{\eta}{\iota} \tag{75}$$

Substitute using $c(T) = c^* e^{-(\rho+\iota)T}$:

$$0 = \frac{c^*}{\rho}e^{-(\rho+\iota)T} - \frac{i}{\iota}\left(e^{-(\rho+\iota)T}\right)^{\frac{\iota}{\rho+\iota}} + \frac{\eta}{\iota}$$

$$\frac{i}{\iota}e^{-\iota T} - \frac{\eta}{\iota} = \frac{c^*}{\rho}e^{-(\rho+\iota)T}$$

$$\frac{i}{\iota}e^{\rho T} - \frac{\eta}{\iota}e^{(\rho+\iota)T} = \frac{c^*}{\rho}$$

and using the optimal reset equation $c^* = \rho(1 + m^*)$ implies

$$\frac{i}{\iota}e^{\rho T} - \frac{\eta}{\iota}e^{(\rho+\iota)T} = m^* + 1$$

■

Proof of Lemma 2.

The inverse consumption function $m(c)$ is given by equation (70). To show that $c(m)$ is unique, it suffices to show that $m(c)$ is strictly increasing on $[c_T, \bar{c}]$, so that the inverse exists and is unique.

From Lemma 6, the derivative of $m(c)$ is

$$m'(c) = \frac{1}{\rho} - \frac{i}{\rho+\iota}\frac{c^{-\rho/(\rho+\iota)}}{(c^*)^{\iota/(\rho+\iota)}}$$

Evaluate at the lower boundary $c = c_T$. Boundary condition (75) implies

$$\frac{1}{c_T}\left(\frac{c_T}{c^*}\right)^{\iota/(\rho+\iota)} = \frac{\iota}{i\rho} + \frac{\eta}{ic_T}$$

so

$$m'(c_T) = \frac{1 - \eta/c_T}{\rho+\iota} > 0$$

Since $c^{-\rho/(\rho+\iota)}$ is strictly decreasing in c, $m'(c)$ is strictly increasing in c. Therefore $m'(c) > m'(c_T) > 0$ for all $c > c_T$, so $m(c)$ is strictly increasing on $[c_T, \bar{c}]$ and $c(m)$ is its well-defined inverse.

The expression for $m'(c)$ also gives $m'(c) < 1/\rho$. By the inverse function theorem,

$$c_0'(m) = \frac{1}{m'(c)} > \rho$$

Differentiating $m'(c)$ gives

$$m''(c) = \frac{i\rho}{(\rho+\iota)^2} \frac{c^{-(2\rho+\iota)/(\rho+\iota)}}{(c^*)^{\iota/(\rho+\iota)}} > 0$$

and therefore

$$c_0''(m) = -\frac{m''(c)}{m'(c)^3} < 0$$

The optimal reset condition gives $c^* = \rho(1+m^*)$. Since $\bar{m} > m^*$,

$$\bar{c} - c^* = \int_{m^*}^{\bar{m}} c'(m)dm > \rho(\bar{m} - m^*)$$

and therefore $\bar{c} > \rho(1+\bar{m})$. Finally, along the riskless policy,

$$\mu(m) = \dot{m} = m'(c)\dot{c} = -(\rho+\iota)c(m)m'(c) < 0$$

This gives $\mu(m) < 0$ for $m \in [0, \bar{m})$. Taking the limit as $m \uparrow \bar{m}$ gives $\mu(\bar{m}) < 0$.

■

Proof of Lemma 7. The optimal reset condition (15) gives

$$c^* = \rho(1+m^*)$$

Lower value matching (16) and the first-order condition $v'(m) = 1/c(m)$ give

$$\frac{1}{\rho}\log(1+m^*) + \varkappa = v(m^*) - v(0) = \int_0^{m^*} \frac{1}{c(m)} dm$$

which is equation (71). ■ **Proof of Lemma 8.**

When the household transfers money into illiquid assets and moves from $\bar{m}$ to the reset target m^*, illiquid assets grow to $A' = (1+\bar{m})A/(1+m^*)$ at cost $\varkappa$. Using $V(mA, A) = \frac{1}{\rho}\log A + v(m)$, the value of this adjustment is

$$V(m^*, A') - \varkappa = \frac{1}{\rho}\log A + \frac{1}{\rho}\log\frac{1+\bar{m}}{1+m^*} + v(m^*) - \varkappa$$

Value-matching at $\bar{m}$ gives

$$v(\bar{m}) - v(m^*) = \frac{1}{\rho}\log\frac{1+\bar{m}}{1+m^*} - \varkappa$$

For the left side, write $v(\bar{m}) - v(m^*) = \int_{m^*}^{\bar{m}} v'(m)\,dm$ and change variables to c, similar to the proof of Lemma 7. Lemma 2 implies that as m increases from m^* to $\bar{m}$, $c(m)$ increases from c^* to $\bar{c}$, with $dm = m'(c)\,dc$ and $v'(m) = 1/c(m)$. Using $m'(c)$ from Lemma 6,

$$\int_{m^*}^{\bar{m}} v'(m)\,dm = \int_{c^*}^{\bar{c}} \frac{m'(c)}{c}\,dc = \int_{c^*}^{\bar{c}} \frac{1}{c}\left(\frac{1}{\rho} - \frac{i}{\rho+\iota}\frac{c^{-\rho/(\rho+\iota)}}{(c^*)^{\iota/(\rho+\iota)}}\right)dc$$

$$= \left(\frac{1}{\rho}\log c + \frac{i}{\rho}\frac{c^{-\rho/(\rho+\iota)}}{(c^*)^{\iota/(\rho+\iota)}}\right)\Bigg|_{c^*}^{\bar{c}} = \frac{1}{\rho}\log\frac{\bar{c}}{c^*} + \frac{i}{\rho}\left(\frac{\bar{c}^{-\rho/(\rho+\iota)}}{(c^*)^{\iota/(\rho+\iota)}} - \frac{1}{c^*}\right)$$

Equating this with the right side gives equation (72). ■

A.4 Proof of Proposition 2

Proof. Set $\eta = 0$, so $\alpha = r$ and $\iota = i$. Lemma 6 gives the paths for consumption and money and the inverse consumption function. Its equation for the reset time becomes

$$m^* = e^{\rho T} - 1$$

which gives the expression for T. The optimal reset condition follows from equation (15).

Using $m'(c)$ to change variables in equation (71) gives

$$\int_0^{m^*} \frac{1}{c(m)} dm = \frac{\rho + i}{\rho^2} \log(1 + m^*) - \frac{i}{\rho^2} \frac{m^*}{1 + m^*}$$

Subtracting $\rho^{-1} \log(1 + m^*)$ gives the relation between $\varkappa$ and m^*. Its derivative with respect to m^* is $im^*/[\rho^2(1 + m^*)^2] > 0$.

The expression for $m'(c)$ is strictly less than $1/\rho$ and strictly increasing in c. At the zero-balance boundary, it equals $1/(\rho + i) > 0$, so the inverse function theorem gives $c'(m) = 1/m'(c) > \rho$. Because $m'(c)$ is increasing in c and c is increasing in m, $c'(m)$ is strictly decreasing in m. ■

A.5 Proof of Corollary 2

Proof. Set $\eta = 0$, so $\iota = i$. Per Lemma 1, the MPC equals $c'(m)$, and by the inverse function theorem $c'(m) = 1/m'(c)$. From Proposition 2, the derivative of the inverse consumption function is

$$m'(c) = \frac{1}{\rho} - \frac{i}{\rho + i} \frac{c^{-\rho/(\rho+i)}}{(c^*)^{i/(\rho+i)}}$$

Evaluating at $c = c^*$ gives

$$m'(c^*) = \frac{1}{\rho} - \frac{i}{\rho + i} \frac{1}{c^*}$$

The optimal reset condition (15) gives $c^* = \rho(1 + m^*)$, so $\frac{1}{c^*} = \frac{1}{\rho(1+m^*)}$. Substituting,

$$m'(c^*) = \frac{1}{\rho} \left(1 - \frac{i}{\rho + i} \frac{1}{1 + m^*} \right)$$

Inverting and using $s^* \equiv \frac{m^*}{m^*+1}$, so that $\frac{1}{1+m^*} = 1 - s^*$,

$$c'(m^*) = \frac{\rho}{1 - \frac{i}{\rho+i}(1 - s^*)} = \frac{\rho(\rho + i)}{\rho + is^*}$$

Since $s^* \in [0, 1)$ and $i > 0$, the denominator satisfies $\rho + is^* < \rho + i$, so $c'(m^*) > \rho$. By Lemma 1, $C_M(M^*, A) = c'(m^*)$, giving the result. ■

A.6 Proof of Corollary 3

Proof. Let $\bar{g}(m)$ denote the stationary density of m, and let $\bar{h}(t)$ denote the stationary density of time since reset. The latter distribution is uniform, so $\bar{h}(t) = 1/T$. Because $m(t)$ is strictly decreasing, its CDF is

$$F_m(x) = \Pr(m(t) \leq x) = \int_{t(x)}^{T} \bar{h}(\tau) d\tau = 1 - \frac{t(x)}{T}$$

Therefore

$$\bar{g}(m) = F'_m(m) = -\frac{1}{T}\frac{dt}{dm}$$

thus the average MPC is

$$\overline{MPC} = \int_0^{m^*} c'(m)\bar{g}(m)dm = -\int_T^0 c'(m(t))\frac{1}{T}dt = \frac{1}{T}\int_0^T c'(m(t))dt$$

Proposition 2 implies

$$c(t) = \rho(1+m^*)e^{-(\rho+i)t} \qquad m(t) = (1+m^*)e^{-(\rho+i)t} - e^{-it}$$

The ratio of time derivatives gives the MPC along the path:

$$c'(m(t)) = \frac{\dot{c}(t)}{\dot{m}(t)} = \frac{\rho(\rho+i)(1+m^*)e^{-\rho t}}{(\rho+i)(1+m^*)e^{-\rho t} - i}$$

Let $q = (1+m^*)e^{-\rho t}$; q falls from $1+m^*$ to 1 between resets, so

$$\int_0^T c'(m(t))dt = \int_1^{1+m^*} \frac{\rho+i}{(\rho+i)q - i}dq = \log\left(1 + \frac{\rho+i}{\rho}m^*\right)$$

Proposition 2 also gives

$$T = \frac{1}{\rho}\log(1+m^*)$$

thus $\overline{MPC} = \rho\frac{\log\left(1+\frac{\rho+i}{\rho}m^*\right)}{\log(1+m^*)}$. ■

A.7 Proof of Proposition 3

Proof. Proposition 2 shows that $\varkappa \to 0$ if and only if $m^* \to 0$ when $i > 0$. Corollary 3 gives

$$\overline{MPC} = \rho\frac{\log\left(1 + \frac{\rho+i}{\rho}m^*\right)}{\log(1+m^*)}$$

Both the numerator and denominator converge to zero as $m^* \to 0$. L'Hôpital's rule therefore gives

$$\lim_{m^*\to 0}\overline{MPC} = \rho\frac{(\rho+i)/\rho}{1} = \rho + i$$

which exceeds ρ when $i > 0$. ■

A.8 Proof of Proposition 4

Lemma 9 (Riskless iMPC representation) *If $\sigma = \eta = 0$ and $i > 0$, let*

$$\tau(m) \equiv \frac{\log(c(m)/c_T)}{\rho+i}$$

be the time for a household starting from m to reach the lower reset boundary, where $c_T \equiv c(m=0)$. The future-consumption object satisfies

$$\mathcal{U}(m,h) = \begin{cases} e^{-(\rho+\pi)h}c(m), & h < \tau(m), \\ \dfrac{e^{\alpha\tau(m)}}{1+m^*}\mathcal{U}(m^*, h-\tau(m)), & h \geq \tau(m) \end{cases} \tag{76}$$

At horizons that are not reset times,

$$\mathcal{I}(m,h) = \frac{c'(m)}{c(m)}\mathcal{U}(m,h) \tag{77}$$

In particular, before the next reset,

$$\mathcal{I}(m,h) = e^{-(\rho+\pi)h}c'(m)$$

Proof.

Let A_0 be assets at the initial date. Before the next reset, Proposition 2 implies

$$c_h = c(m)e^{-(\rho+i)h}$$

Assets grow at rate α, so $A_h = A_0 e^{\alpha h}$. Hence

$$\mathcal{U}(m,h) = \frac{C_h}{A_0} = e^{\alpha h}c(m)e^{-(\rho+i)h} = e^{-(\rho+\pi)h}c(m)$$

where $i = \pi+\alpha$ because $\eta = 0$. The time to the lower boundary is the time at which $c(m)e^{-(\rho+i)\tau(m)} = c_T$, which gives the stated expression for $\tau(m)$.

If $h \geq \tau(m)$, the household reaches the lower boundary at $\tau(m)$, at which point assets are $A_0 e^{\alpha\tau(m)}$. The reset portfolio condition places the household at m^* with new illiquid assets $A_0 e^{\alpha\tau(m)}/(1+m^*)$. Future consumption per original asset is therefore

$$\mathcal{U}(m,h) = \frac{e^{\alpha\tau(m)}}{1+m^*}\mathcal{U}(m^*, h-\tau(m))$$

which proves the renewal equation.

For the derivative, first note that before the next reset direct differentiation gives

$$\mathcal{I}(m,h) = e^{-(\rho+\pi)h}c'(m) = \frac{c'(m)}{c(m)}\mathcal{U}(m,h)$$

Now consider horizons after the first reset but away from reset times. Write $s = h - \tau(m)$ and $R(m) = e^{\alpha\tau(m)}/(1+m^*)$, so $\mathcal{U}(m,h) = R(m)\mathcal{U}(m^*,s)$. Between reset times, the renewal equation implies $\partial_h \mathcal{U}(m^*,s) = -(\rho+\pi)\mathcal{U}(m^*,s)$. Differentiating gives

$$\partial_m \mathcal{U}(m,h) = R(m)\tau'(m)\left(\alpha+\rho+\pi\right)\mathcal{U}(m^*,s)$$

Since $\alpha+\rho+\pi = \rho+i$ and

$$\tau'(m) = \frac{c'(m)}{(\rho+i)c(m)}$$

this becomes $\partial_m \mathcal{U}(m,h) = [c'(m)/c(m)]\mathcal{U}(m,h)$. Repeating the same argument after subsequent resets proves the all-horizon formula away from reset times.

■

Proof of Proposition 4. Before the next reset, Lemma 9 gives

$$\mathcal{I}(m,h) = e^{-(\rho+\pi)h}c'(m)$$

Taking the log derivative with respect to h gives (20). If $\eta = 0$ and $r = \rho$, then $i = \pi + r = \pi + \rho$, giving $\partial_h \log \mathcal{I}(m,h) = -i$ and $\mathcal{I}(m,h) = e^{-ih}c'(m)$ before the next reset. ■

A.9 Proof of Proposition 12

Proof.

The transfer raises the household's normalized money balance from m_0 to $m_0 + \Delta$. There are two cases for the immediate consumption response, depending on whether the new balance $m_0 + \Delta$ lies inside the inaction region or above the upper boundary $\bar{m}$.

First, if the payment does not cause the household to reset its money balance, normalized consumption jumps from $c(m_0)$ to $c(m_0 + \Delta)$, giving

$$dC = A\left[c(m_0 + \Delta) - c(m_0)\right]$$

In the second case, the household immediately pays the fixed cost and resets. At the moment of reset, total wealth is $(1 + m_0 + \Delta)A$. The optimal reset condition (15) places normalized money at m^*, so the portfolio constraint (7) implies that the new money balance is $M^* = m^*A'$ where the new illiquid assets are $A' = (1 + m_0 + \Delta)A/(1 + m^*)$. Post-reset consumption is

$$C' = c^*A' = \rho(1 + m^*)\frac{(1 + m_0 + \Delta)A}{1 + m^*} = \rho(1 + m_0 + \Delta)A$$

so the change in consumption is

$$dC = \rho(1 + m_0 + \Delta)A - c(m_0)A = A\left(\rho(1 + m_0 + \Delta) - c(m_0)\right)$$

At $\Delta = \bar{m} - m_0$ the two cases give different levels. Lemma 2 gives $\bar{c} > \rho(1 + \bar{m})$, so Case 1 gives $A[\bar{c} - c(m_0)]$ while Case 2 gives $A[\rho(1 + \bar{m}) - c(m_0)] < A[\bar{c} - c(m_0)]$. The response drops discontinuously at the threshold, with a downward jump of $A[\bar{c} - \rho(1 + \bar{m})] > 0$. The slope also drops, from $c'(\bar{m}) > \rho$ (below threshold) to ρ (above threshold).

■

A.10 Proof of Lemma 3

Proof.

Differentiate the optimized HJB (12) with respect to m. The derivative of $\log c(m)$ is $c'(m)/c(m)$; the derivative of $v'(m)\mu_\sigma(m)$ contributes $v''(m)\mu_\sigma(m) + v'(m)(-\iota - c'(m))$. The first-order condition $1/c(m) = v'(m)$ implies the $c'(m)$ terms cancel; rearranging gives (21).

■

A.11 Proof of Proposition 5

Let c_0 denote the riskless solution with the same income allocation η, given by Lemma 2, let $c_0^* \equiv \rho(1 + m_0^*)$, and write $\mu_0(m) \equiv \eta - \iota m - c_0(m)$ and $\lambda_0(m) \equiv 1/c_0(m)$. Define

$$\Theta(m) \equiv \exp\left(\int_{m_0^*}^{m} \frac{\rho + \iota + c_0'(s)}{\mu_0(s)}\, ds\right) \qquad \Upsilon(m) \equiv -\Theta(m)\int_{m_0^*}^{m} \frac{\lambda_0''(s)}{2\mu_0(s)\Theta(s)}\, ds \tag{78}$$

Thus $\Theta(m_0^*) = 1$ and $\Upsilon(m_0^*) = 0$. Also define

$$\Xi \equiv -\frac{\int_0^{m_0^*} \Upsilon(m)\, dm}{\int_0^{m_0^*} \Theta(m)\, dm} \tag{79}$$

Define the gap between the smooth-pasting target for marginal utility and its one-sided riskless value by

$$\Delta\lambda_{\bar{m}} \equiv \frac{1}{\rho(1+\bar{m}_0)} - \frac{1}{\bar{c}_0} > 0$$

We first establish the following lemma.

Lemma 10 (Small-Risk Expansion) *For $\sigma > 0$ sufficiently small, the stochastic consumption function satisfies smooth pasting $c_\sigma(\bar{m}_\sigma) = \rho(1+\bar{m}_\sigma)$. The following hold:*

1. *Marginal utility satisfies*

$$\lambda_\sigma(m) = \lambda_0(m) + \sigma^2\hat{\lambda}(m) + \Delta\lambda_{\bar{m}}\, e^{2\mu_0(\bar{m}_0)(\bar{m}_\sigma - m)/\sigma^2} + \mathcal{R}_\sigma(m) \tag{80}$$

   where

$$\hat{\lambda}(m) = \Xi\,\Theta(m) + \Upsilon(m) \tag{81}$$

   For each fixed $m \in [0, \bar{m}_0)$,

$$\mathcal{R}_\sigma(m) = O(\sigma^4) \qquad \mathcal{R}'_\sigma(m) = O(\sigma^4)$$

   For each fixed $\xi \geq 0$,

$$\mathcal{R}_\sigma(\bar{m}_\sigma - \sigma^2\xi) = O(\sigma^2) \qquad \partial_\xi \mathcal{R}_\sigma(\bar{m}_\sigma - \sigma^2\xi) = O(\sigma^2)$$

2. *Define*

$$\hat{c}(m) \equiv -c_0(m)^2\hat{\lambda}(m) \tag{82}$$

   For each fixed $m \in [0, \bar{m}_0)$,

$$c_\sigma(m) = c_0(m) + \sigma^2\hat{c}(m) + O(\sigma^4) \tag{83}$$

   and

$$c'_\sigma(m) = c'_0(m) + \sigma^2\hat{c}'(m) + O(\sigma^4) \tag{84}$$

3. *The reset threshold satisfies*

$$m_\sigma^* = m_0^* + \sigma^2 \frac{(c_0^*)^2\,\hat{\lambda}(m_0^*)}{c'_0(m_0^*) - \rho} + O(\sigma^4) \tag{85}$$

   where $c'_0(m_0^*) - \rho > 0$ by Lemma 2.

4. *The upper boundary satisfies*

$$\bar{m}_\sigma = \bar{m}_0 + \sigma^2\left[\frac{1}{\Delta\lambda_{\bar{m}}}\int_{m_0^*}^{\bar{m}_0} \hat{\lambda}(m)\, dm - \frac{1}{2\mu_0(\bar{m}_0)}\right] + O(\sigma^4) \tag{86}$$

Proof of Lemma 10.

For each fixed $m < \bar{m}_0$, decompose the risky marginal utility function λ_σ in terms of the zero-order (risk-free) term λ_0, the first-order term $\hat{\lambda}$, and higher-order terms:

$$\lambda_\sigma = \lambda_0 + \sigma^2 \hat{\lambda} + O(\sigma^4)$$

Since $\mu_0(m) < 0$ for each fixed $m < \bar{m}_0$, the coefficients in the envelope equation are smooth at that point. The expansion and its first derivative with respect to m therefore have $O(\sigma^4)$ remainders. Using $c_\sigma = 1/\lambda_\sigma$, the risky consumption function satisfies

$$c_\sigma = c_0 - \sigma^2 c_0^2 \hat{\lambda} + O(\sigma^4)$$

and the risky drift $\mu_\sigma(m) = \eta - \iota m - c_\sigma(m)$ satisfies

$$\mu_\sigma = \mu_0 + \sigma^2 c_0^2 \hat{\lambda} + O(\sigma^4)$$

Substituting into the envelope condition (21) gives

$$(\rho + \iota)\left(\lambda_0 + \sigma^2 \hat{\lambda}\right) = \mu_0 \left(\lambda_0' + \sigma^2 \hat{\lambda}'\right) + \sigma^2 c_0^2 \hat{\lambda} \lambda_0' + \tfrac{\sigma^2}{2}\left(\lambda_0'' + \sigma^2 \hat{\lambda}''\right) + O(\sigma^4)$$

The coefficients on σ^2 imply the ODE

$$(\rho + \iota)\hat{\lambda} = \mu_0 \hat{\lambda}' + c_0^2 \lambda_0' \hat{\lambda} + \tfrac{1}{2}\lambda_0''$$

and, using $\lambda_0' = -c_0'/c_0^2$, equivalently

$$\hat{\lambda}'\, \mu_0 - (\rho + \iota + c_0')\, \hat{\lambda} = -\tfrac{1}{2}\lambda_0'' \tag{87}$$

which is first order in $\hat{\lambda}$. The function $\hat{\lambda}$ defined by (81) satisfies (87). From (78),

$$\frac{\Theta'(m)}{\Theta(m)} = \frac{\rho + \iota + c_0'(m)}{\mu_0(m)}, \qquad \Upsilon'(m) = \frac{\rho + \iota + c_0'(m)}{\mu_0(m)}\Upsilon(m) - \frac{\lambda_0''(m)}{2\mu_0(m)}.$$

Differentiating $\hat{\lambda} = \Xi\Theta + \Upsilon$ therefore gives $\hat{\lambda}' = \frac{\rho+\iota+c_0'}{\mu_0}\hat{\lambda} - \lambda_0''/(2\mu_0)$; multiplying by μ_0 recovers (87).

Next we apply the perturbation method to the boundary conditions. Decompose the optimal boundaries as

$$m_\sigma^* = m_0^* + \sigma^2 \hat{m}^* + O(\sigma^4) \qquad\qquad \bar{m}_\sigma = \bar{m}_0 + \sigma^2 \hat{\bar{m}} + O(\sigma^4)$$

Apply the expansions to write lower value-matching (16) as:

$$\int_0^{m_0^*} (\lambda_0 + \sigma^2 \hat{\lambda})\, dm + \int_{m_0^*}^{m_0^* + \sigma^2 \hat{m}^*} (\lambda_0 + \sigma^2 \hat{\lambda})\, dm + O(\sigma^4) = \varkappa + \tfrac{1}{\rho}\log(1 + m_0^*) + \frac{1}{\rho(1 + m_0^*)}\sigma^2 \hat{m}^*$$

The short integral evaluates as $\int_{m_0^*}^{m_0^* + \sigma^2 \hat{m}^*} \lambda_0\, dm = \lambda_0(m_0^*) \cdot \sigma^2 \hat{m}^* + O(\sigma^4)$, since the interval has length $O(\sigma^2)$ (mean-value theorem for integrals). Thus:

$$\int_0^{m_0^*} (\lambda_0 + \sigma^2 \hat{\lambda})\, dm + \sigma^2 \hat{m}^* \lambda_0(m_0^*) + O(\sigma^4) = \varkappa + \tfrac{1}{\rho}\log(1 + m_0^*) + \frac{1}{\rho(1 + m_0^*)}\sigma^2 \hat{m}^*$$

Collect coefficients on σ^2:

$$\int_0^{m_0^*} \hat{\lambda}\, dm + \hat{m}^* \lambda_0(m_0^*) = \frac{1}{\rho(1 + m_0^*)}\hat{m}^* \implies \int_0^{m_0^*} \hat{\lambda}\, dm = 0$$

because the optimal reset condition implies $\lambda_0(m_0^*) = \frac{1}{\rho(1+m_0^*)}$. The integral condition $\int_0^{m_0^*} \hat{\lambda}\, dm = 0$ holds by the choice of Ξ in (79): substituting (81) gives $\Xi \int_0^{m_0^*} \Theta\, dm + \int_0^{m_0^*} \Upsilon\, dm = 0$. For each fixed $m < \bar{m}_0$, $c_\sigma = 1/\lambda_\sigma$ gives

$$c_\sigma(m) = c_0(m) + \sigma^2 \hat{c}(m) + O(\sigma^4)$$

and differentiating,

$$c'_\sigma(m) = c'_0(m) + \sigma^2 \hat{c}'(m) + O(\sigma^4)$$

These match (83) and (84). The marginal-utility approximation extending to the upper boundary is established below.

To derive (85), expand reset optimality (15):

$$\lambda_0(m_0^*) + \sigma^2 \hat{\lambda}(m_0^*) + \lambda'_0(m_0^*)\sigma^2 \hat{m}^* + O(\sigma^4) = \frac{1}{\rho(1+m_0^*)} - \frac{1}{\rho(1+m_0^*)^2}\sigma^2 \hat{m}^*$$

Collect coefficients on σ^2 then use that $\lambda'_0 = -c'_0/c_0^2$ and $c_0^* = \rho(1+m_0^*)$:

$$\hat{\lambda}(m_0^*) + \lambda'_0(m_0^*)\hat{m}^* = -\frac{1}{\rho(1+m_0^*)^2}\hat{m}^*$$

$$\hat{\lambda}(m_0^*) - \frac{c'_0(m_0^*)}{(c_0^*)^2}\hat{m}^* = -\frac{\rho}{(c_0^*)^2}\hat{m}^*$$

$$(c_0^*)^2 \hat{\lambda}(m_0^*) - c'_0(m_0^*)\, \hat{m}^* = -\rho\, \hat{m}^*$$

$$\implies \hat{m}^* = \frac{(c_0^*)^2\, \hat{\lambda}(m_0^*)}{c'_0(m_0^*) - \rho}$$

For (80) and (86), the interior expansion predicts $\lambda_\sigma(\bar{m}_0) \approx 1/\bar{c}_0$, but the smooth-pasting condition (iii) requires $\lambda_\sigma(\bar{m}_\sigma) \approx 1/(\rho(1+\bar{m}_0))$. The gap $\Delta\lambda_{\bar{m}} = O(1)$ cannot be closed by adding further interior terms; it requires a separate boundary-layer analysis covering an $O(\sigma^2)$ neighborhood of $\bar{m}_\sigma$.

Introduce $\xi = (\bar{m}_\sigma - m)/\sigma^2$, so $\xi = 0$ is the upper boundary and $\xi \to \infty$ is the interior. Define $\Lambda(\xi) \equiv \lambda_\sigma(\bar{m}_\sigma - \sigma^2\xi)$. By the chain rule, $\lambda'_\sigma = -\Lambda'(\xi)/\sigma^2$ and $\lambda''_\sigma = \Lambda''(\xi)/\sigma^4$. Substituting into (21) and multiplying through by σ^2:

$$\sigma^2(\rho + \iota)\Lambda = -\mu_\sigma(\bar{m}_\sigma - \sigma^2\xi)\, \Lambda' + \tfrac{1}{2}\Lambda''$$

For each fixed ξ, the point $m = \bar{m}_\sigma - \sigma^2\xi$ is within $O(\sigma^2)$ of $\bar{m}_0$, so $\mu_\sigma(\bar{m}_\sigma - \sigma^2\xi) = \mu_0(\bar{m}_0) + O(\sigma^2)$. Dropping the $O(\sigma^2)$ terms gives the boundary-layer ODE

$$\tfrac{1}{2}\Lambda'' - \mu_0(\bar{m}_0)\, \Lambda' = 0$$

The characteristic equation $\frac{1}{2}r^2 - \mu_0(\bar{m}_0)\, r = 0$ has roots $r = 0$ and $r = 2\mu_0(\bar{m}_0)$, giving the general solution $\Lambda(\xi) = A + Be^{2\mu_0(\bar{m}_0)\xi}$; since $\mu_0(\bar{m}_0) < 0$, the second term decays as $\xi \to \infty$. Matching as $\xi \to \infty$ to the interior expansion gives $A = \lambda_0(\bar{m}_0) = 1/\bar{c}_0$. At $\xi = 0$, smooth-pasting requires $\Lambda(0) = 1/(\rho(1+\bar{m}_\sigma)) = 1/(\rho(1+\bar{m}_0)) + O(\sigma^2)$, so $A + B = 1/(\rho(1+\bar{m}_0))$ and $B = \Delta\lambda_{\bar{m}}$. The composite formula (80) follows by adding the interior expansion and this boundary-layer solution, then subtracting their shared limit $1/\bar{c}_0$ to avoid double-counting. The coefficients in the equation for Λ are smooth in σ^2. The terms omitted from that equation are $O(\sigma^2)$, so both $\mathcal{R}_\sigma(\bar{m}_\sigma - \sigma^2\xi)$ and its derivative with respect to ξ are $O(\sigma^2)$ for each fixed ξ. For each fixed $m < \bar{m}_0$, the exponential is smaller than any power of σ, and the interior construction gives $\mathcal{R}_\sigma(m) = O(\sigma^4)$ and $\mathcal{R}'_\sigma(m) = O(\sigma^4)$. In that interior, $c_\sigma = 1/\lambda_\sigma$ gives (83) and (84). Differentiating the composite approximation and using $c'_\sigma = -(\lambda_\sigma^{\text{comp}})'/(\lambda_\sigma^{\text{comp}})^2$ shows that the exponential contributes $[2\mu_0(\bar{m}_0)/\sigma^2]\Delta\lambda_{\bar{m}}\, e^{2\mu_0(\bar{m}_0)(\bar{m}_\sigma - m)/\sigma^2}/(\lambda_\sigma^{\text{comp}})^2$ to c'_σ. This term is negative

and of order σ^{-2} near $\bar{m}_\sigma$, so consumption falls over an interval of width $O(\sigma^2)$ immediately below the upper boundary.

For (86), write $\bar{m}_\sigma = \bar{m}_0 + \sigma^2\hat{\bar{m}} + O(\sigma^4)$ and expand upper value-matching (17) as

$$-\int_{m_\sigma^*}^{\bar{m}_\sigma} \lambda_\sigma \, dm = \varkappa + \tfrac{1}{\rho}\log\tfrac{1+m_\sigma^*}{1+\bar{m}_\sigma}$$

The full λ_σ consists of the interior component $\lambda_0+\sigma^2\hat{\lambda}+O(\sigma^4)$ plus the boundary-layer exponential.

For the interior component, expand the limits using the Leibniz rule:

$$-\int_{m_\sigma^*}^{\bar{m}_\sigma} (\lambda_0 + \sigma^2\hat{\lambda} + O(\sigma^4))\, dm = -\int_{m_0^*}^{\bar{m}_0} (\lambda_0 + \sigma^2\hat{\lambda})\, dm - \sigma^2\hat{\bar{m}}\lambda_0(\bar{m}_0) + \sigma^2\hat{m}^*\lambda_0(m_0^*) + O(\sigma^4)$$

For the boundary-layer component, change variables $\xi = (\bar{m}_\sigma - m)/\sigma^2$, $dm = -\sigma^2\, d\xi$; as m runs from m_σ^* to $\bar{m}_\sigma$, ξ runs from ∞ to 0:

$$-\int_{m_\sigma^*}^{\bar{m}_\sigma} \Delta\lambda_{\bar{m}} e^{2\mu_0(\bar{m}_0)\xi}\, dm = -\sigma^2\int_0^\infty \Delta\lambda_{\bar{m}} e^{2\mu_0(\bar{m}_0)\xi}\, d\xi = \frac{\sigma^2\Delta\lambda_{\bar{m}}}{2\mu_0(\bar{m}_0)} + O(\sigma^4)$$

because $\mu_0(\bar{m}_0) < 0$. Combining, the full LHS integral is:

$$-\int_{m_\sigma^*}^{\bar{m}_\sigma} \lambda_\sigma\, dm = -\int_{m_0^*}^{\bar{m}_0} (\lambda_0 + \sigma^2\hat{\lambda})\, dm - \sigma^2\hat{\bar{m}}\lambda_0(\bar{m}_0) + \sigma^2\hat{m}^*\lambda_0(m_0^*) + \frac{\sigma^2\Delta\lambda_{\bar{m}}}{2\mu_0(\bar{m}_0)} + O(\sigma^4)$$

Expanding the RHS as well and collecting all σ^2 coefficients:

$$-\int_{m_0^*}^{\bar{m}_0} \hat{\lambda}\, dm - \hat{\bar{m}}\lambda_0(\bar{m}_0) + \hat{m}^*\lambda_0(m_0^*) + \frac{\Delta\lambda_{\bar{m}}}{2\mu_0(\bar{m}_0)} = \frac{\hat{m}^*}{\rho(1+m_0^*)} - \frac{\hat{\bar{m}}}{\rho(1+\bar{m}_0)}$$

Since $\lambda_0(m_0^*) = 1/(\rho(1+m_0^*))$, the $\hat{m}^*$ terms cancel exactly, as with the optimal reset condition. Rearrange the $\hat{\bar{m}}$ terms and use $1/\bar{c}_0 - 1/(\rho(1+\bar{m}_0)) = -\Delta\lambda_{\bar{m}}$ to get

$$-\int_{m_0^*}^{\bar{m}_0} \hat{\lambda}\, dm + \hat{\bar{m}}\Delta\lambda_{\bar{m}} + \frac{\Delta\lambda_{\bar{m}}}{2\mu_0(\bar{m}_0)} = 0 \implies \hat{\bar{m}} = \frac{1}{\Delta\lambda_{\bar{m}}}\int_{m_0^*}^{\bar{m}_0} \hat{\lambda}\, dm - \frac{1}{2\mu_0(\bar{m}_0)}$$

In the interior, the exponential term is negligible and the expansion satisfies the envelope ODE (21) to $O(\sigma^4)$ by construction of $\hat{\lambda}$ from (87). Near $\bar{m}_\sigma$, the pointwise remainder is $O(\sigma^2)$. Since this region has width $O(\sigma^2)$, the omitted terms contribute $O(\sigma^4)$ to upper value-matching. Thus the threshold expansions (85) and (86) retain their $O(\sigma^4)$ remainders. Lower value-matching is satisfied by the choice of Ξ in (79), optimal reset by (85), smooth-pasting to the stated order by $B = \Delta\lambda_{\bar{m}}$, and upper value-matching by (86).

■

Proof of Proposition 5.

Equations (81) and (82) give the closed-form expression for $\hat{c}$. Equations (83) and (84) give the two approximations for each fixed $m \in [0, \bar{m}_0)$. Equations (85) and (86) give

$$m_\sigma^* = m_0^* + O(\sigma^2) \qquad \bar{m}_\sigma = \bar{m}_0 + O(\sigma^2)$$

Lemma 10 also gives $c_\sigma(\bar{m}_\sigma) = \rho(1+\bar{m}_\sigma)$.

For the behavior near the upper boundary, let $m = \bar{m}_\sigma - \sigma^2\xi$. Equation (80) gives, to leading

order,

$$\lambda_\sigma(\bar{m}_\sigma - \sigma^2\xi) = \frac{1}{\bar{c}_0} + \Delta\lambda_{\bar{m}} e^{2\mu_0(\bar{m}_0)\xi}$$

As ξ falls toward zero, consumption falls from approximately $\bar{c}_0$ to $\rho(1+\bar{m}_\sigma)$. A fixed interval in ξ has width $O(\sigma^2)$ in m. Differentiating the exponential in (80) gives the leading term in $c'_\sigma(m)$ near the upper boundary:

$$\frac{2\mu_0(\bar{m}_0)}{\sigma^2}\frac{\Delta\lambda_{\bar{m}} e^{2\mu_0(\bar{m}_0)(\bar{m}_\sigma - m)/\sigma^2}}{\lambda_\sigma(m)^2}$$

This term is of order σ^{-2} and is negative because $\mu_0(\bar{m}_0) < 0$ and $\Delta\lambda_{\bar{m}} > 0$. All remaining terms are $O(1)$ on this scale by Lemma 10. Therefore $c'_\sigma(m) < 0$ close to $\bar{m}_\sigma$ for σ sufficiently small.

■

A.12 Proof of Theorem 1

Proof. Property 1 follows directly from Lemma 1: the MPC equals $c'(m)$, which depends only on m.

For property 2, equation (84) gives $c'_\sigma(m) = c'_0(m) + O(\sigma^2)$ uniformly on $[0, m_0^*]$. Since $c'_0(m) > \rho$ for all $m \in [0, m_0^*]$ by Lemma 2 and $[0, m_0^*]$ is compact, there exists $\bar{\sigma} > 0$ such that $c'_\sigma(m) > \rho$ on $[0, m_\sigma^*]$ for all $\sigma < \bar{\sigma}$.

For property 3, in the interior ($m \leq \bar{m}_0 - \varepsilon$ for any fixed $\varepsilon > 0$), equation (84) gives $c''_\sigma = c''_0 + O(\sigma^2) < 0$ for small σ, since $c''_0 < 0$ by Lemma 2 and the $O(\sigma^2)$ term is uniformly bounded on compacta. In the boundary layer near $\bar{m}_\sigma$, differentiating the composite (80) shows that c'_σ is dominated by the term $\frac{2\mu_0(\bar{m}_0)}{\sigma^2}\Delta\lambda_{\bar{m}}\, e^{2\mu_0(\bar{m}_0)(\bar{m}_\sigma - m)/\sigma^2}/\lambda_\sigma^2$; since $\mu_0(\bar{m}_0) < 0$, this term is negative and decreasing in m, so c'_σ continues to fall.

For property 4, smooth pasting forces $c_\sigma(\bar{m}_\sigma) = \rho(1+\bar{m}_\sigma)$. Since $\bar{c}_0 > \rho(1+\bar{m}_0)$ by Lemma 2 and $c_\sigma \to c_0$ uniformly on compact sets, c_σ must decrease sharply from $c_0(\bar{m}_0 - \varepsilon) \approx \bar{c}_0$ to $\rho(1+\bar{m}_\sigma)$ within the boundary layer. The composite formula (80) shows this transition occurs over an interval of width $O(\sigma^2)$, so $c'_\sigma < 0$ throughout this layer. ■

A.13 Proof of Proposition 6

Proof.

Fix current assets A_0 and normalized money m. In the inaction region, normalized money follows the diffusion

$$dm = \mu_\sigma(m)dt - \sigma dW^q,$$

while assets grow deterministically as $dA = \alpha A\, dt$. Over a short interval dh, expected future consumption per current asset therefore satisfies the dynamic programming identity

$$\mathcal{U}_\sigma(m, h+dh) = e^{\alpha dh}\mathbb{E}_m[\mathcal{U}_\sigma(m_{dh}, h)] + o(dh).$$

Expanding the right side and using the generator

$$\mathcal{L}_\sigma f(m) \equiv \mu_\sigma(m) f'(m) + \frac{\sigma^2}{2} f''(m)$$

gives

$$\partial_h \mathcal{U}_\sigma = \alpha\mathcal{U}_\sigma + \mathcal{L}_\sigma\mathcal{U}_\sigma,$$

which is (22). The initial condition follows because horizon-zero consumption per current asset is $c_\sigma(m)$.

At the lower boundary, the household immediately resets from $m = 0$ to m^*_σ. If the pre-reset asset level is A, the new illiquid asset level is $A/(1 + m^*_\sigma)$ by the portfolio constraint, so future consumption per pre-reset asset is $\mathcal{U}_\sigma(m^*_\sigma, h)/(1 + m^*_\sigma)$. This gives (23). At the upper boundary, total wealth is $(1 + \bar{m}_\sigma)A$, so after reset the new illiquid asset level is $(1 + \bar{m}_\sigma)A/(1 + m^*_\sigma)$, which gives (24).

Differentiating the initial condition in m gives $\mathcal{I}_\sigma(m, 0) = c'_\sigma(m)$. Finally, away from boundary layers and reset times, the diffusion with drift μ_σ converges to the deterministic riskless flow as $\sigma \to 0$, while Proposition 5 gives $c_\sigma \to c_0$ and $m^*_\sigma \to m^*_0$. Standard continuous-dependence of the backward equation on its coefficients then gives $\mathcal{U}_\sigma \to \mathcal{U}$ and, on compact sets where the limiting solution is smooth, $\partial_m \mathcal{U}_\sigma \to \partial_m \mathcal{U}$.

■

A.14 Proof of Lemma 4

Proof of Lemma 4. For a twice continuously differentiable function z on $[0, \bar{m}]$, define

$$\mathbf{Z}(t) \equiv \int_0^{\bar{m}} z(m)\mathcal{A}(m, t)\, dm$$

This is the aggregate of $Az(m)$, and finite aggregate wealth makes the integral finite. Inside the inaction region, normalized money and illiquid assets follow

$$dm = \mu(m)\, dt - \sigma\, dW^q \qquad dA = \alpha A\, dt$$

Itô's lemma and the product rule imply

$$\begin{aligned} d[Az(m)] &= z(m)dA + Az'(m)dm + \frac{A}{2}z''(m)(dm)^2 \\ &= A\left[\alpha z(m) + \mu(m)z'(m) + \frac{\sigma^2}{2}z''(m)\right] dt - \sigma A z'(m)\, dW^q \end{aligned}$$

The drift $\mu(m) = \eta - (\pi + \alpha)m - c(m)$ includes liquid income, inflation, consumption, and the decline in normalized money caused by asset growth. The diffusion term includes the stochastic change in normalized money, while $\alpha z(m)$ captures the growth in the asset weight.

At a reset from boundary $b \in \{0, \bar{m}\}$, a household with illiquid assets A moves from (b, A) to $(m^*, \psi_b A)$. Its change in $Az(m)$ is

$$A\left[\psi_b z(m^*) - z(b)\right]$$

Taking expectations and aggregating the continuous changes and reset changes gives

$$\begin{aligned} \dot{\mathbf{Z}}(t) = &\int_0^{\bar{m}} \left[\alpha z(m) + \mu(m)z'(m) + \frac{\sigma^2}{2}z''(m)\right] \mathcal{A}(m, t)\, dm \\ &+ \left[\psi_0 z(m^*) - z(0)\right] \mathbf{F}^A_0(t) \\ &+ \left[\psi_{\bar{m}} z(m^*) - z(\bar{m})\right] \mathbf{F}^A_{\bar{m}}(t) \end{aligned} \tag{88}$$

Equations (26) and (27), together with $\bar{M}(A) = \bar{m}A$, imply

$$\tilde{G}(0, A, t) = 0 \qquad \tilde{G}(\bar{m}, A, t) = 0$$

Equation (32) therefore gives the boundary conditions in the lemma.

Multiplying the right side of equation (38) by $z(m)$ and integrating by parts gives the integral

term in equation (88). The boundary terms are

$$\frac{\sigma^2}{2}\left[z\partial_m\mathcal{A} - z'\mathcal{A}\right]_0^{\bar{m}} - \left[z\mu\mathcal{A}\right]_0^{\bar{m}} = -z(0)\mathbf{F}_0^A(t) - z(\bar{m})\mathbf{F}_{\bar{m}}^A(t)$$

using the boundary conditions and equation (37). The reset source contribution is

$$z(m^*)\left[\psi_0\mathbf{F}_0^A(t) + \psi_{\bar{m}}\mathbf{F}_{\bar{m}}^A(t)\right]$$

Together, these expressions give equation (88) for every z. This determines the law of motion for $\mathcal{A}$, giving equation (38).

Finally, equations (28) and (32) give

$$\mathcal{A}(m,0) = \int_0^\infty A^2\Psi(mA, A)\,dA$$

■

A.15 Proof of Proposition 7

Proof. We apply equation (88) from the proof of Lemma 4 in Appendix A.14 to assets, money, and consumption.

First set $z(m) = 1$. Since $\mathbf{Z} = \mathbf{A}$, equation (88) gives

$$\dot{\mathbf{A}} = \alpha\mathbf{A} + (\psi_0 - 1)\mathbf{F}_0^A + (\psi_{\bar{m}} - 1)\mathbf{F}_{\bar{m}}^A$$

Using equation (40), this is equation (43).

Next set $z(m) = m$. Portfolio conservation implies

$$\psi_b(1 + m^*) = 1 + b \qquad \Longrightarrow \qquad \psi_b m^* - b = 1 - \psi_b \qquad b \in \{0, \bar{m}\}$$

Moreover,

$$\alpha m + \mu(m) = \eta - \pi m - c(m)$$

Since $\mathbf{Z} = \mathbf{M}$, substituting these identities and equation (40) into equation (88) gives equation (44).

Finally set $z(m) = c(m)$. The envelope equation (21) and the first-order condition $c = 1/\lambda$ imply

$$\mu(m)c'(m) + \frac{\sigma^2}{2}c''(m) = -(\rho + \iota)c(m) + \sigma^2\frac{c'(m)^2}{c(m)}$$

Using $\iota = \pi + \alpha$, the interior term in equation (88) becomes

$$-(\rho + \pi)\mathbf{C}(t) + \sigma^2\mathbf{P}(t)$$

The optimal reset condition and upper smooth-pasting condition imply

$$\begin{aligned}\psi_0 c(m^*) - c(0) &= \rho - c(0)\\ \psi_{\bar{m}} c(m^*) - c(\bar{m}) &= 0\end{aligned}$$

Substituting the interior and reset terms gives equation (45). ■

A.16 Proof of Lemma 5

Proof. Until the first reset, $A(t) = A(0)e^{\alpha t}$ and $Z = m^* A(0)$. If $b \in \{0, \bar{m}\}$ is the boundary reached at that reset, then

$$\mathcal{F}_B(t \mid x) = \mathbb{E}\left[e^{\alpha\tau}\psi_b \mathbf{1}\{\tau \le t\} \,|\, m(0) = m^* x\right]$$
$$\mathcal{F}_J(t \mid x) = \frac{1}{m^*}\mathbb{E}\left[e^{\alpha\tau}(1-\psi_b)\mathbf{1}\{\tau \le t\} \,|\, m(0) = m^* x\right]$$

The scale $A(0)$ cancels. At every reset, the process restarts at m^*.

Decompose each marginal flow according to the order of the reset after the transfer. Let $\mathbf{B}_k^{\varepsilon_x}(t)$ and $\mathbf{L}_k^{\varepsilon_x}(t)$ denote the contributions from reset k. The transfer changes the initial relative state to $x+\varepsilon_x$ while leaving Z fixed. Differentiating the first-reset functions and integrating against $\bar{h}_Z$ gives

$$\mathbf{B}_1^{\varepsilon_x}(t) = D_B(t) \qquad \mathbf{L}_1^{\varepsilon_x}(t) = D_J(t) \tag{89}$$

After reset $k-1$ at date s, a household begins a new spell at $x = 1$ with scale equal to its post-reset balance. Homogeneity makes future flows linear in this scale. Thus $\mathbf{B}_{k-1}^{\varepsilon_x}(s)ds$ units of marginal scale entering at date s generate the reset-k flows

$$\mathbf{B}_k^{\varepsilon_x}(t) = \int_0^t f_B^*(t-s)\mathbf{B}_{k-1}^{\varepsilon_x}(s)\,ds$$
$$\mathbf{L}_k^{\varepsilon_x}(t) = \int_0^t f_J^*(t-s)\mathbf{B}_{k-1}^{\varepsilon_x}(s)\,ds \tag{90}$$

for $k \ge 2$. Growth within the new spell is already included in f_B^* and f_J^* through $e^{\alpha\tau}$. The reset process is nonexplosive, so summing over reset orders and interchanging the sum and integral gives

$$\begin{aligned}
\mathbf{B}^{\varepsilon_x}(t) &= D_B(t) + \int_0^t f_B^*(t-s)\sum_{k=2}^{\infty}\mathbf{B}_{k-1}^{\varepsilon_x}(s)\,ds \\
&= D_B(t) + \int_0^t f_B^*(t-s)\mathbf{B}^{\varepsilon_x}(s)\,ds \\
\mathbf{L}^{\varepsilon_x}(t) &= D_J(t) + \int_0^t f_J^*(t-s)\sum_{k=2}^{\infty}\mathbf{B}_{k-1}^{\varepsilon_x}(s)\,ds \\
&= D_J(t) + \int_0^t f_J^*(t-s)\mathbf{B}^{\varepsilon_x}(s)\,ds
\end{aligned} \tag{91}$$

where $\mathbf{B}^{\varepsilon_x} = \sum_{k\ge1}\mathbf{B}_k^{\varepsilon_x}$ and $\mathbf{L}^{\varepsilon_x} = \sum_{k\ge1}\mathbf{L}_k^{\varepsilon_x}$.

■

A.17 Proof of Proposition 8

Proof. Let A_k^- and A_k^+ denote illiquid assets immediately before and after adjustment k. Growth between adjustments and wealth preservation at the reset give

$$A_k^- = A_{k-1}^+ e^{\alpha\tau_k} \qquad A_k^+ = A_k^- - \Delta M_k$$

Substituting the first identity into the second, multiplying by m^*, and using $B_k = m^* A_k^+$ gives equation (64). ■

B Generality of the Functional Sufficient Statistic

The functional sufficient statistic is not specific to the consumption function $c(m)$ or the (s, S) adjustment policies derived in our household model. This appendix introduces a more general KFE and shows that it gives the same aggregate budget equations and reset-flow equations used in Section 6. The extensions change the inputs to the sufficient statistic while preserving its form.

Consdier the following KFE describing the asset-weighted measure $\mathcal{A}(m,t)$:

$$\partial_t \mathcal{A} = \frac{1}{2}\partial_m^2\big(\sigma(m)^2\mathcal{A}\big) - \partial_m(\mu(m)\mathcal{A}) + (\alpha - \zeta(m))\mathcal{A} + \delta(m - m^*)\frac{\mathbf{B}(t)}{m^*} \tag{92}$$

Equation (38) from Section 5.4 is a special case. The drift $\mu(m) = \eta - (\pi + \alpha)m - c(m)$ allows an arbitrary normalized consumption function $c(m)$. The function $\sigma(m)$ allows risk to depend on balances, and $\zeta(m) \geq 0$ is an additional adjustment hazard inside the inaction region. For example, in Alvarez and Lippi (2009) random low-cost withdrawal opportunities induce households to replenish cash before exhausting their balances. The boundaries satisfy $0 \leq \underline{m} < m^* < \bar{m} < \infty$, allowing a positive lower adjustment threshold as in Lippi and Moracci (2026).

We maintain the assumptions that consumption is given by $C = Ac(m)$, assets growth at rate α between adjustments, and normalized money resets to some m^*. Assume that $c \geq 0$ and $\sigma > 0$ are twice continuously differentiable on $[\underline{m}, \bar{m}]$, that ζ is continuous, and that aggregate assets are finite. Households adjust immediately at either boundary, so the boundary conditions remain

$$\mathcal{A}(\underline{m}, t) = \mathcal{A}(\bar{m}, t) = 0$$

The outward boundary flows, measured in pre-adjustment assets per unit of time, are

$$\mathbf{F}^A_{\underline{m}}(t) = \frac{\sigma(\underline{m})^2}{2}\partial_m \mathcal{A}(\underline{m}, t) \qquad \mathbf{F}^A_{\bar{m}}(t) = -\frac{\sigma(\bar{m})^2}{2}\partial_m \mathcal{A}(\bar{m}, t)$$

An adjustment from m multiplies assets by $\psi(m) \equiv (1+m)/(1+m^*)$. Writing $\psi_b = \psi(b)$ at either boundary, the net liquidation and post-reset balance flows become

$$\begin{aligned} \mathbf{L}(t) &= \sum_{b \in \{\underline{m}, \bar{m}\}} (1 - \psi_b)\mathbf{F}^A_b(t) + \int_{\underline{m}}^{\bar{m}} (1 - \psi(m))\zeta(m)\mathcal{A}(m,t)\, dm \\ \mathbf{B}(t) &= m^* \left[\sum_{b \in \{\underline{m}, \bar{m}\}} \psi_b \mathbf{F}^A_b(t) + \int_{\underline{m}}^{\bar{m}} \psi(m)\zeta(m)\mathcal{A}(m,t)\, dm \right] \end{aligned} \tag{93}$$

These definitions include both boundary and interior adjustments. Consider the same transfer $\varepsilon_m A$ as in Section 6, starting from a stationary normalized asset-weighted density and holding the coefficients, returns, and reset rule fixed.

Proposition 9 *The functional sufficient statistic in Theorem 3 and the reset-flow equations in Lemma 5 hold under the generalized KFE* (92)*, with the stationary distribution, consumption responses, and adjustment functions generated by these dynamics. The MPC aggregation result in Theorem 2 also holds.*

Proof. Integrating the KFE against 1 and m gives the same aggregate budget equations (43)–(44):

$$\dot{\mathbf{A}} = \alpha\mathbf{A} - \mathbf{L} \qquad \dot{\mathbf{M}} = \eta\mathbf{A} - \pi\mathbf{M} - \mathbf{C} + \mathbf{L}$$

The interior diffusion terms vanish in both calculations. The drift satisfies $\mu(m) + \alpha m = \eta - \pi m - c(m)$, and wealth preservation gives $m^*\psi(m) - m = 1 - \psi(m)$. Thus boundary and interior

adjustments contribute $-\mathbf{L}$ to assets and $+\mathbf{L}$ to money, including the outgoing money at a positive lower boundary.

The reset-flow equations use the common reset point and the scaling of household balances. Let τ be the first adjustment time and m_{τ^-} the normalized balance immediately before adjustment. Until τ, assets grow as $A(t) = A(0)e^{\alpha t}$. With initial scale $Z = m^*A(0)$ and relative balance $x = m(0)/m^*$, the next-reset functions are

$$\mathcal{F}_B(t \mid x) = \mathbb{E}\left[e^{\alpha\tau}\psi(m_{\tau^-})\mathbf{1}\{\tau \le t\} \mid m(0) = m^*x\right]$$
$$\mathcal{F}_J(t \mid x) = \frac{1}{m^*}\mathbb{E}\left[e^{\alpha\tau}(1-\psi(m_{\tau^-}))\mathbf{1}\{\tau \le t\} \mid m(0) = m^*x\right]$$

Summing over successive adjustments therefore gives the renewal equations in Lemma 5, by the argument in Appendix A.16. The bounded hazard and interior reset point ensure that the same uniqueness argument applies.

The proportional transfer shifts m by ε_m, so its aggregate consumption response is still the asset-weighted average iMPC. Because the initial density vanishes at both boundaries, there is no first-order adjustments on impact. Differentiating the aggregate budget equations therefore yields Theorem 3, with the same initial conditions.

Finally, the joint density vanishes at $M = \underline{m}A$ and $M = \bar{m}A$. The boundary term in the integration-by-parts proof of Theorem 2 is consequently zero, so the aggregate MPC from an equal-dollar transfer still equals the number-weighted average individual MPC. ■

C Power Utility

This appendix generalizes the utility function from our simple baseline model and demonstrates the main results hold: the MPC is distorted in the cashless limit, and the iMPC slope depends on inflation. Moreover, these distortions are increasing in the elasticity of intertemporal substitution (EIS).

Separability of the value function depends on homotheticity. Log utility is the simplest example, but we now assume the more general power utility:

$$u(C) = \frac{C^{1-\gamma}}{1-\gamma} \qquad \gamma > 0, \quad \gamma \neq 1$$

and let the utility cost of an adjustment be $\varkappa(M+A)^{1-\gamma}$. This cost is independent of the amount transferred, but scales with total wealth to preserve the normalization in Section 2.3. Otherwise we retain the assumed economic environment and stochastic processes. And households follow the same adjustment policy: households reset to m^* when normalized money reaches either 0 or $\bar{m}$, with $0 < m^* < \bar{m} < \infty$.

C.1 Normalization

We use the same normalization as in Section 2.3: $m \equiv M/A$ and $c \equiv C/A$. Homogeneity implies

$$V(M,A) = A^{1-\gamma}v(m) \qquad C(M,A) = Ac(m)$$

Using $\alpha = r - \eta$ and $\iota = \pi + \alpha$, the normalized HJB becomes

$$[\rho - (1-\gamma)\alpha]v(m) = \max_{c>0}\left\{\frac{c^{1-\gamma}}{1-\gamma} + v'(m)(\eta - \iota m - c) + \frac{\sigma^2}{2}v''(m)\right\} \tag{94}$$

The first-order condition is $v'(m) = c(m)^{-\gamma}$, and the MPC remains $C_M(M,A) = c'(m)$.

At a reset, the household preserves total wealth, so assets become $A(1+m)/(1+m^*)$. Choosing the reset balance to maximize the post-reset value gives

$$(1+m^*)v'(m^*) = (1-\gamma)v(m^*) \tag{95}$$

Value matching at either boundary requires

$$v(b) = \left(\frac{1+b}{1+m^*}\right)^{1-\gamma} v(m^*) - \varkappa(1+b)^{1-\gamma} \qquad b \in \{0, \bar{m}\} \tag{96}$$

For $\sigma > 0$, optimal adjustment at the upper boundary requires smooth pasting:

$$(1+\bar{m})v'(\bar{m}) = (1-\gamma)v(\bar{m})$$

Smooth pasting reveals one noticeable difference from the log-utility case: because the adjustment cost now varies with wealth, equating the marginal values of continuing and adjusting at the boundary does not imply equating consumption, which may jump discontinuously.

C.2 A Deterministic Case

As in Section 3, we illustrate the central result with the simple case where $\sigma = \eta = 0$. We also require $\rho > (1-\gamma)r$, so that frictionless consumption is positive, and $i > 0$ so that money is costly to hold. The frictionless consumption rule is

$$C(M, A) = \frac{\rho + (\gamma - 1)r}{\gamma}(M + A)$$

and the frictionless MPC is therefore $[\rho + (\gamma-1)r]/\gamma$, which equals ρ when $r = \rho$.

Proposition 10 *Normalized consumption and money evolve after a reset according to*

$$c(t) = c^* e^{-\left(r + \frac{\rho+\pi}{\gamma}\right)t} \tag{97}$$

$$m(t) = e^{-it}\left(m^* - c^* \int_0^t e^{-(\rho+(1-\gamma)\pi)s/\gamma}\, ds\right) \tag{98}$$

where $t \in [0, T]$ is time since the reset and $c(t) \equiv c(m(t))$. Normalized consumption at the optimal reset point is

$$c^* = \frac{\rho + (\gamma-1)r}{\gamma} + \frac{\rho + (1-\gamma)\pi}{\gamma} m^* \tag{99}$$

the time between resets T satisfies

$$m^* = c^* \int_0^T e^{-(\rho+(1-\gamma)\pi)t/\gamma}\, dt \tag{100}$$

and the MPC is strictly decreasing in m on $[0, m^*]$ and satisfies

$$c'(m) = \frac{\left(r + \frac{\rho+\pi}{\gamma}\right) c(m)}{im + c(m)} \qquad c'(0) = \frac{\rho + (\gamma-1)r}{\gamma} + \frac{i}{\gamma} \tag{101}$$

Proof. Differentiating the HJB (94) with respect to m and using $v' = c^{-\gamma}$ gives

$$\dot{c} = -\left(r + \frac{\rho + \pi}{\gamma}\right) c \tag{102}$$

$$\dot{m} = -im - c \tag{103}$$

Solving equation (102) gives the consumption path (97). Substituting this path into equation (103) and solving gives the money path (98). Substitute the reset condition (95) into the HJB at m^* to yield equation (99). Setting $m(T) = 0$ in equation (98) gives the duration equation (100), and $c'(m) = \dot{c}/\dot{m}$ implies equation (101). Along the path,

$$c(t) - \frac{\rho + (1-\gamma)\pi}{\gamma} m(t) = \frac{\rho + (\gamma - 1)r}{\gamma} e^{-it} > 0$$

Differentiating (101) implies $c''(m) < 0$. ■

Consumption smoothing again raises the MPC near zero balances, with a gap of i/γ from the frictionless MPC at the zero-balance boundary. If $0 < \gamma < 1$, the MPC exceeds the frictionless benchmark throughout the cycle. For $\gamma > 1$, the MPC is elevated during the entire period between resets if and only if

$$m^* < \frac{\rho + (\gamma - 1)r}{(\gamma - 1)(\rho + \pi + \gamma r)}$$

which follows from evaluating equation (101) at m^*, where the MPC is lowest. It holds for sufficiently small adjustment costs, but possibly not for large costs.

C.3 The Cashless Limit is not Cashless

Small adjustment costs shrink the range of normalized money balances between resets without eliminating the MPC distortion. As in Corollary 3, let $\overline{MPC}$ denote the average MPC between resets.

Proposition 11 (The Cashless Limit under Power Utility) *Under the deterministic assumptions above,*

$$\lim_{\varkappa \to 0} \overline{MPC} = \lim_{m^* \to 0} \overline{MPC} = r + \frac{\rho + \pi}{\gamma} = \frac{\rho + (\gamma - 1)r}{\gamma} + \frac{i}{\gamma}$$

Proof. Let $c_T \equiv \lim_{m \downarrow 0} c(m) = c^* e^{-[r + (\rho + \pi)/\gamma]T}$ denote normalized consumption just before liquidation. The HJB at zero, the optimal reset condition, and value matching imply

$$\varkappa = \frac{1}{1-\gamma}\left[\left(\frac{1+m^*}{c^*}\right)^{\gamma} - \frac{\gamma c_T^{1-\gamma}}{\rho + (\gamma - 1)r}\right]$$

Differentiating after substituting for c^* and T shows that this cost is strictly increasing in $m^* > 0$ wherever $c^* > 0$. Expanding at zero gives

$$\varkappa = \frac{i}{2}\left(\frac{\gamma}{\rho + (\gamma - 1)r}\right)^{\gamma + 1} (m^*)^2 + O((m^*)^3)$$

Thus $\varkappa \to 0$ implies $m^* \to 0$. The MPCs at 0 and m^* in (101) then converge to $r + (\rho + \pi)/\gamma$. Since the MPC is decreasing in balances, its average over the cycle has the same limit. ■

The intuition is the same as the log utility case. When $r = \rho$, Proposition 10 pins the boundary MPC at $c'(0) = \rho + i/\gamma$; sending the adjustment cost to zero implies that average normalized money balances approach zero, and thus the average MPC approaches $c'(0)$. As before, the Friedman rule

($i = 0$) eliminates the distortion. The new insight provided by power utility is that a greater EIS $\frac{1}{\gamma}$ increases the distortion.

C.4 Intertemporal MPCs

Consumption in levels evolves between resets according to

$$C(t) = C(0)e^{-(\rho+\pi)t/\gamma}$$

As in Proposition 4, we hold initial assets fixed and differentiate with respect to initial money. Before the next reset,

$$\mathcal{I}(m,h) = e^{-(\rho+\pi)h/\gamma}c'(m) \qquad \partial_h \log \mathcal{I}(m,h) = -\frac{\rho+\pi}{\gamma}$$

When $r = \rho$, the iMPC therefore declines at rate i/γ. Again, the main findings hold qualitatively, but are affected by power utility: increasing the EIS also increases the effect of the inflation rate on the iMPC slope.

D The Consumption Effects of Large Payments

Small payments affect consumption through the distorted MPC $c'(m)$. For larger payments, the consumption response depends on both the payment size and the household's money balance at the time of the payment, as Proposition 12 shows.

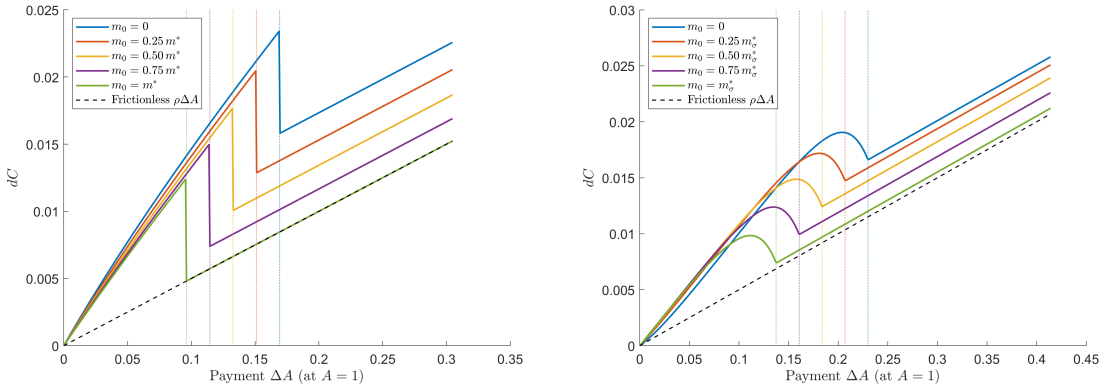
Proposition 12 *If all income is illiquid ($\eta = 0$, $\alpha = r$), the steady state condition holds ($\rho = r$), and a household with normalized money balance $m_0 \in [0, m^*]$ receives a lump-sum monetary transfer ΔA at time 0, then their immediate change in consumption is*

$$dC(m_0,\Delta) = \begin{cases} A\left[c(m_0+\Delta) - c(m_0)\right] & \text{if } m_0+\Delta \leq \bar{m} \\ A\left[\rho(1+m_0+\Delta) - c(m_0)\right] & \text{if } m_0+\Delta > \bar{m} \end{cases} \tag{104}$$

At the threshold $\Delta = \bar{m} - m_0$ the response drops discontinuously: the level falls by $A[\bar{c} - \rho(1+\bar{m})] > 0$ (Lemma 2), and the slope drops from $c'(\bar{m}) > \rho$ to ρ.

Proof: Appendix A.9.

Figure 7a illustrates the nonlinearity and state-dependence of Proposition 12, plotting the consumption change dC (with $A = 1$) against the payment size Δ for households at different initial positions m_0. For small Δ, the response exceeds the frictionless benchmark $\rho\Delta$ because $c'(m) > \rho$ everywhere in the inaction region. As Δ grows, it eventually pushes the household past the reset threshold $\bar{m}$, after which the household resets and consumes $C = \rho(A + M)$. This creates a discrete drop in the response function at $\Delta = \bar{m} - m_0$: because smooth pasting fails in the riskless model ($\bar{c} > \rho(1+\bar{m})$, Lemma 2), pre-reset consumption exceeds post-reset consumption. Above the threshold the MPC collapses to exactly ρ. The right panel shows the same picture for the risky model at the calibrated value of σ. The qualitative shape is the same—responses exceed the frictionless benchmark for small Δ, and collapse to ρ above the threshold—but the discrete drop disappears: smooth pasting holds in the risky model ($c_\sigma(\bar{m}_\sigma) = \rho(1+\bar{m}_\sigma)$, Proposition 5), so the response joins the frictionless line continuously.



(a) Riskless model

(b) Risky model

Figure 7: Consumption Response to a Monetary Payment

Note: Immediate change in consumption dC (at $A = 1$) following a lump-sum monetary payment ΔA (Proposition 12), for households at different initial money positions m_0. *Left (riskless):* Below $\bar{m}$, the response is nonlinear and exceeds the frictionless benchmark $\rho\Delta A$. At the reset threshold $\bar{m} - m_0$ (dotted vertical lines), the response drops discontinuously because smooth pasting fails at $\bar{m}$ ($\bar{c} > \rho(1 + \bar{m})$, Lemma 2). Above the threshold the MPC equals exactly ρ. *Right (risky):* Same structure at the calibrated value of σ, but the response is *continuous* at the reset threshold because smooth pasting holds: $c_\sigma(\bar{m}_\sigma) = \rho(1 + \bar{m}_\sigma)$ (Proposition 5). The $O(\sigma^2)$ composite approximation is used for c_σ. Dashed line: frictionless benchmark $\rho\Delta A$.

E MPCs Under Income Risk

E.1 The MPC at Zero Balances

Proposition 13 *If $\eta = 0$ and $\sigma > 0$ is small, risk lowers the MPC at the zero-balance boundary. Specifically,*

$$c'_\sigma(0) = \rho + i - \sigma^2 \frac{(\rho + i)(2\rho + 3i)}{2\rho^2}(1 + m_0^*)^{2i/\rho} + O(\sigma^4)$$

Proof. Equation (84) gives

$$c'_\sigma(0) = c'_0(0) + \sigma^2 \hat{c}'(0) + O(\sigma^4)$$

Let $c_T \equiv c_0(0)$. When $\eta = 0$, $\mu_0(0) = -c_T$ and Corollary 1 gives $c'_0(0) = \rho + i$. Differentiating the riskless envelope condition at $m = 0$ gives

$$c''_0(0) = -\frac{i(\rho + i)}{c_T}$$

Since $\lambda_0 = 1/c_0$,

$$\lambda''_0(0) = \frac{2c'_0(0)^2}{c_T^3} - \frac{c''_0(0)}{c_T^2} = \frac{(\rho + i)(2\rho + 3i)}{c_T^3}$$

At $m = 0$, equation (87) becomes

$$-c_T \hat{\lambda}'(0) - 2(\rho + i)\hat{\lambda}(0) = -\frac{1}{2}\lambda''_0(0)$$

Using $\hat{c} = -c_0^2 \hat{\lambda}$ therefore gives

$$\hat{c}'(0) = -2c_T(\rho + i)\hat{\lambda}(0) - c_T^2 \hat{\lambda}'(0) = -\frac{c_T}{2}\lambda''_0(0) = -\frac{(\rho + i)(2\rho + 3i)}{2c_T^2}$$

Finally, Proposition 2 implies

$$c_T = \rho(1+m_0^*)e^{-(\rho+i)\log(1+m_0^*)/\rho} = \rho(1+m_0^*)^{-i/\rho}$$

Substituting for c_T in the $\hat{c}'(0)$ expression gives the result. ■

E.2 The iMPC Horizon Elasticity

Let $m_0(s;m)$ denote the normalized money balance at horizon s for a household beginning at m in the riskless model. Define

$$\Gamma_0(m) \equiv \left(\frac{c_0'(m)}{c_0(m)}\right)^2 \left[1 + \frac{2(c_0'(m)-\rho)}{\rho+\iota}\right]$$

Proposition 14 *For each fixed $m \in (0, \bar{m}_0)$, each horizon h strictly before the riskless household reaches the lower boundary, and $\sigma > 0$ sufficiently small, income risk makes the iMPC decay more quickly than in the riskless model. Specifically, $\Gamma_0(m) > 0$ and*

$$\partial_h \log \mathcal{I}_\sigma(m,h) = -(\rho+\pi) - \sigma^2\Gamma_0(m_0(h;m)) + O(\sigma^4) \tag{105}$$

Proof. The envelope condition (21) and $c_\sigma = 1/\lambda_\sigma$ imply

$$\mu_\sigma(m)c_\sigma'(m) + \frac{\sigma^2}{2}c_\sigma''(m) = -(\rho+\iota)c_\sigma(m) + \sigma^2\frac{c_\sigma'(m)^2}{c_\sigma(m)}$$

Between resets, Itô's lemma and $\iota = \pi + \alpha$ therefore give

$$dC_t = \left[-(\rho+\pi)C_t + \sigma^2 A_t \frac{c_\sigma'(m_t)^2}{c_\sigma(m_t)}\right]dt - \sigma A_t c_\sigma'(m_t)dW_t^q$$

Because h is strictly before the riskless household reaches the lower boundary, the path $m_0(s;m)$ remains a fixed positive distance from both boundaries for $s \in [0,h]$. A risky household can reset before h only if its path crosses this distance. For some $K > 0$, this event has probability $O(e^{-K/\sigma^2})$, so resets do not affect the expansion through order σ^4.

Multiplying the consumption equation by $e^{(\rho+\pi)t}$ gives

$$d\left(e^{(\rho+\pi)t}C_t\right) = \sigma^2 e^{(\rho+\pi)t}A_t\frac{c_\sigma'(m_t)^2}{c_\sigma(m_t)}dt - \sigma e^{(\rho+\pi)t}A_t c_\sigma'(m_t)dW_t^q$$

Integrating from 0 to h gives

$$e^{(\rho+\pi)h}C_h = C_0 + \sigma^2\int_0^h e^{(\rho+\pi)s}A_s\frac{c_\sigma'(m_s)^2}{c_\sigma(m_s)}\,ds - \sigma\int_0^h e^{(\rho+\pi)s}A_s c_\sigma'(m_s)\,dW_s^q$$

The stochastic integral has expectation zero. Since $A_s = A_0e^{\alpha s}$, dividing by A_0 and taking expectations gives

$$\mathcal{U}_\sigma(m,h) = e^{-(\rho+\pi)h}\left[c_\sigma(m) + \sigma^2\int_0^h e^{(\rho+\pi+\alpha)s}\mathbb{E}_m\left[\frac{c_\sigma'(m_s)^2}{c_\sigma(m_s)}\right]ds\right]$$

To first order in σ, write

$$m_s = m_0(s;m) + \sigma z_s + O(\sigma^2)$$

where z_s is defined as the coefficient on σ in the expansion. The normalized money equation (14)

implies

$$dz_s = \mu_0'(m_0(s;m))z_s\,ds - dW_s^q \qquad z_0 = 0$$

and therefore $\mathbb{E}_m[z_s] = 0$. Proposition 5 and a second-order expansion now give

$$\mathbb{E}_m\left[\frac{c_\sigma'(m_s)^2}{c_\sigma(m_s)}\right] = \frac{c_0'(m_0(s;m))^2}{c_0(m_0(s;m))} + O(\sigma^2)$$

Using $\rho + \pi + \alpha = \rho + \iota$ and equation (83) now gives

$$\mathcal{U}_\sigma(m,h) = e^{-(\rho+\pi)h}\left[c_0(m) + \sigma^2\hat{c}(m) + \sigma^2\int_0^h e^{(\rho+\iota)s}\frac{c_0'(m_0(s;m))^2}{c_0(m_0(s;m))}\,ds\right] + O(\sigma^4)$$

Along the riskless path,

$$c_0(m_0(s;m)) = e^{-(\rho+\iota)s}c_0(m)$$

and therefore

$$\partial_m m_0(s;m) = e^{-(\rho+\iota)s}\frac{c_0'(m)}{c_0'(m_0(s;m))}$$

The riskless envelope condition implies

$$\mu_0(m)c_0'(m) = -(\rho+\iota)c_0(m)$$

Differentiating this equation and using $\mu_0'(m) = -\iota - c_0'(m)$ gives

$$\mu_0(m)c_0''(m) = (c_0'(m) - \rho)c_0'(m)$$

The first equation also gives $\mu_0(m) = -(\rho+\iota)c_0(m)/c_0'(m)$. Substituting this into the second equation gives

$$\frac{c_0''(m)}{c_0(m)} = -\left(\frac{c_0'(m)}{c_0(m)}\right)^2\frac{c_0'(m)-\rho}{\rho+\iota}$$

Therefore, by the definition of Γ_0,

$$\frac{2c_0''(m)}{c_0(m)} - \left(\frac{c_0'(m)}{c_0(m)}\right)^2 = -\Gamma_0(m)$$

Differentiating $\mathcal{U}_\sigma(m,h)$ with respect to m and using equation (84) gives

$$\mathcal{I}_\sigma(m,h) = e^{-(\rho+\pi)h}c_\sigma'(m)\left[1 - \sigma^2\int_0^h \Gamma_0(m_0(s;m))\,ds\right] + O(\sigma^4)$$

Taking the log derivative with respect to h gives equation (105).

Lemma 2 gives $c_0'(m) > \rho$, so $\Gamma_0(m) > 0$. ■

F An Approximation Approach for the KFE

This appendix constructs an approximate solution for the distribution of normalized money $g(m,t)$. We first state the Green's-function solution for a general drift, then apply it to a linear approximation of the consumption function. Throughout this appendix, $\sigma > 0$.

Let $\mu(m)$ denote the normalized drift, m^* the reset point, and $\bar{m}$ the upper boundary. Let ϕ be the initial probability measure, possibly with interior point masses. Conditional on the reset

frequency $F(t)$, the normalized KFE is

$$\partial_t g(m,t) = \frac{\sigma^2}{2}\partial_{mm} g(m,t) - \partial_m(\mu(m)g(m,t)) + \delta(m-m^*)F(t) \tag{106}$$

with

$$g(0,t) = g(\bar{m},t) = 0 \qquad g(m,0)\,dm = \phi(dm)$$

Conservation of households implies

$$F(t) = \frac{\sigma^2}{2}\left[\partial_m g(0,t) - \partial_m g(\bar{m},t)\right] \tag{107}$$

F.1 The Green's Function

Let $\mathcal{G}(m,y,t)$ denote the Green's function for equation (106) when $F = 0$. A Green's function is a distributional IRF: $\mathcal{G}(m,y,t)$ gives the density over m at horizon t generated by a unit mass initially located at y, removed when it reaches either boundary.

The conditional survival function and reset-time density are

$$S(t \mid y) \equiv \int_0^{\bar{m}} \mathcal{G}(m,y,t)\,dm \tag{108}$$

$$\begin{aligned} f(t \mid y) &\equiv -\partial_t S(t \mid y) \\ &= \frac{\sigma^2}{2}\left[\partial_m \mathcal{G}(0,y,t) - \partial_m \mathcal{G}(\bar{m},y,t)\right] \end{aligned} \tag{109}$$

Property 1 *Define*

$$a_\phi(t) \equiv \int_0^{\bar{m}} f(t \mid y)\,\phi(dy) \qquad b(t) \equiv f(t \mid m^*)$$

The reset frequency satisfies

$$F(t) = a_\phi(t) + \int_0^t F(\tau) b(t-\tau)\,d\tau \qquad \widehat{F}(s) = \frac{\widehat{a}_\phi(s)}{1-\widehat{b}(s)} \tag{110}$$

where hats denote Laplace transforms and the second expression holds for $\mathrm{Re}(s) > 0$. *The dynamic distribution is*

$$g(m,t) = \int_0^{\bar{m}} \mathcal{G}(m,y,t)\,\phi(dy) + \int_0^t F(\tau)\mathcal{G}(m,m^*,t-\tau)\,d\tau \tag{111}$$

The function a_ϕ captures first resets from the initial distribution, while b captures first resets from households beginning a new spell at m^*. Thus equation (110) determines F without first solving for g. The distribution then follows from equation (111): the first term propagates the initial distribution, while the second adds the distributions generated by households that reset at each earlier date. This property follows from Property 1 and Theorem 1 of Adams (2026). Moreover, $|\widehat{b}(s)| < 1$ for $\mathrm{Re}(s) > 0$ because b is the density of a strictly positive, finite first-reset time.

Corollary 4 *Define the mean time between resets as*

$$\bar{T} \equiv \int_0^\infty S(t \mid m^*)\,dt = \int_0^{\bar{m}} \int_0^\infty \mathcal{G}(m,m^*,t)\,dt\,dm \tag{112}$$

The long-run reset frequency and stationary distribution are

$$\bar{F} = \frac{1}{\bar{T}} \qquad \bar{g}(m) = \bar{F} \int_0^\infty \mathcal{G}(m, m^*, t)\, dt \tag{113}$$

Proof. The first-reset time has a continuous density and finite mean, so equation (110) implies that the reset frequency converges to $\bar{F}$. In the long run, the first term in equation (111) disappears. Taking the limit gives the expression for $\bar{g}$, and integrating it over m gives $\bar{F}\bar{T} = 1$. ■

Thus the stationary distribution records the time spent at each m during a spell beginning at m^*, normalized to integrate to one. The long-run reset frequency is the inverse of the expected spell length.

F.2 A Linear Approximation

Let m^* and $\bar{m}$ denote the riskless reset point and upper boundary, and let $c_T = c(0)$. When adjustment costs are small, the inaction region is narrow. We use the line through the reset value $c^* = \rho(1 + m^*)$ with slope equal to the riskless MPC at the lower boundary:

$$\tilde{c}(m) = c^* + s_0(m - m^*) \qquad s_0 \equiv \frac{(\rho + \iota)c_T}{c_T - \eta} \tag{114}$$

The corresponding drift is

$$\tilde{\mu}(m) \equiv \eta - \iota m - \tilde{c}(m) = \beta_0 - \beta_1 m \qquad \beta_0 \equiv \eta - c^* + s_0 m^* \qquad \beta_1 \equiv \iota + s_0 \tag{115}$$

When $\eta = 0$, these expressions reduce to $s_0 = \rho + i$, $\beta_0 = im^* - \rho$, and $\beta_1 = \rho + 2i$. At the baseline calibration, $\beta_0 < 0$, so the approximate drift points toward the lower boundary throughout the inaction region.

Small adjustment costs imply that the riskless consumption function differs from equation (114) by $O(\bar{m}^2)$. Proposition 5 implies that the risky policy differs from the riskless policy by $O(\sigma^2)$ away from the $O(\sigma^2)$ interval immediately below the upper boundary. Therefore

$$\mu_\sigma(m) = \eta - \iota m - c_\sigma(m) = \tilde{\mu}(m) + O(\bar{m}^2 + \sigma^2)$$

away from that interval. Proposition 5 also implies that replacing the risky reset point and upper boundary with m^* and $\bar{m}$ changes them by $O(\sigma^2)$. This is an approximation to the drift; it is not a uniform error bound for the distribution.

For the money model, $\mu = \mu_\sigma$. Replacing it in equation (106) with $\tilde{\mu}$ gives the approximate KFE

$$\partial_t g(m,t) = \frac{\sigma^2}{2}\partial_{mm} g(m,t) - \partial_m((\beta_0 - \beta_1 m)g(m,t)) + \delta(m - m^*)F(t) \tag{116}$$

with the boundary and initial conditions given above.

F.3 The Linear Green's Function

Define

$$Q(m) \equiv \int_0^m \tilde{\mu}(s)\, ds = \beta_0 m - \frac{\beta_1}{2} m^2 \tag{117}$$

Lemma 11 *Let $(\lambda_n, \varphi_n)_{n\geq 1}$ solve*

$$\frac{\sigma^2}{2}\varphi_n''(m) - \left(\frac{\tilde{\mu}'(m)}{2} + \frac{\tilde{\mu}(m)^2}{2\sigma^2}\right)\varphi_n(m) = -\lambda_n \varphi_n(m) \tag{118}$$

with $0 < \lambda_1 < \lambda_2 < \cdots$, $\varphi_n(0) = \varphi_n(\bar{m}) = 0$, *and*

$$\int_0^{\bar{m}} \varphi_n(m)\varphi_j(m)\, dm = \mathbf{1}\{n = j\}$$

The Green's function for the linear approximation is

$$\mathcal{G}_{\text{lin}}(m, y, t) = e^{[Q(m)-Q(y)]/\sigma^2} \sum_{n=1}^{\infty} \varphi_n(m)\varphi_n(y)e^{-\lambda_n t} \tag{119}$$

Proof. Write the Green's function as $\mathcal{G}_{\text{lin}}(m, y, t) = e^{Q(m)/\sigma^2} u(m, y, t)$. Because $Q' = \tilde{\mu}$, the KFE for u is generated by the operator on the left-hand side of equation (118). Its initial condition is $u(m, y, 0) = e^{-Q(y)/\sigma^2}\delta(m - y)$, so its eigenfunction expansion gives equation (119).

For computation, define

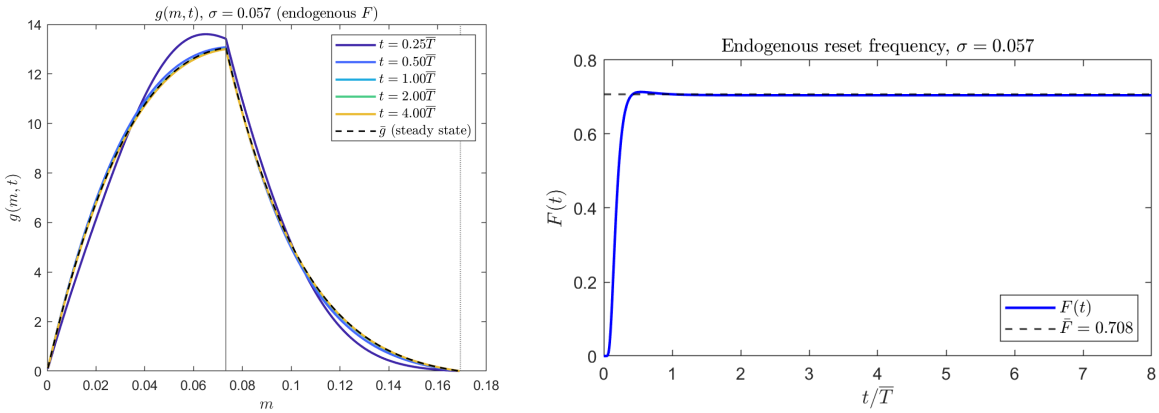
$$z \equiv \sqrt{\frac{2\beta_1}{\sigma^2}} \left(m - \frac{\beta_0}{\beta_1} \right) \qquad \nu_n \equiv \frac{\lambda_n}{\beta_1}$$

Then equation (118) becomes

$$\frac{d^2\varphi_n}{dz^2} + \left(\nu_n + \frac{1}{2} - \frac{z^2}{4} \right) \varphi_n = 0 \tag{120}$$

Its solutions are the parabolic cylinder functions $U(-\nu_n - 1/2, z)$ and $V(-\nu_n - 1/2, z)$. The two boundary conditions determine the eigenvalues. ■

For the point-mass initial condition in Figure 8, $a_\phi = b$ and $\widehat{F} = \widehat{b}/(1 - \widehat{b})$. The figure applies Property 1 under the linear approximation.



(a) Dynamic distribution

(b) Reset frequency

Figure 8: Dynamic Distribution and Reset Frequency Under the Linear Approximation

Note: The initial distribution is a point mass at m^* and parameters take their calibrated values. Left: the dynamic distribution at several horizons; the dashed line is the stationary distribution, while the solid and dotted vertical lines mark m^* and $\bar{m}$. Right: the endogenous reset frequency determined by equation (110); the dashed line is the long-run frequency from Corollary 4.

The stationary objects can also be calculated without solving the eigenvalue problem. Define

$$\mathcal{L}(m) \equiv e^{2Q(m)/\sigma^2} \int_0^m e^{-2Q(s)/\sigma^2}\, ds \qquad \mathcal{R}(m) \equiv e^{2Q(m)/\sigma^2} \int_m^{\bar{m}} e^{-2Q(s)/\sigma^2}\, ds \tag{121}$$

Then

$$\bar{g}(m) = C_g \begin{cases} \mathcal{L}(m) & 0 \le m < m^* \\ \dfrac{\mathcal{L}(m^*)}{\mathcal{R}(m^*)} \mathcal{R}(m) & m^* < m \le \bar{m} \end{cases} \qquad \bar{F} = \frac{\sigma^2 C_g}{2} \left(1 + \frac{\mathcal{L}(m^*)}{\mathcal{R}(m^*)} \right) \tag{122}$$

where C_g is chosen so that $\int_0^{\bar{m}} \bar{g}(m)\, dm = 1$. This is equivalent to Corollary 4. Because Q is quadratic, the two integrals reduce to incomplete Gaussian integrals.

G Parameter Sensitivity

Which parameters govern liquidity and the speed of adjustment? Table 3 reports numerical partial derivatives around the baseline in Table 2. Each entry is the change in a model statistic per unit increase in the row parameter, holding the other parameters fixed. The parameter coordinates are $(\rho, i, r, \eta, \sigma, \varkappa)$, with $\pi = i - r$. Thus a change in i holds r fixed, while a change in r holds i fixed and changes inflation in the opposite direction. We vary ρ independently of r and do not recalibrate to preserve the baseline moments.

Table 3: Local Sensitivity of Model Statistics

Parameter	$\mathbf{M/A}$	R_Δ	$h_{1/2}$	F_L	F_I	m^*	$\bar{m}$	$\overline{MPC}$
ρ	0.691	3.42	263	-6.65	-4.3	1.71	4.18	0.444
i	-0.278	-0.125	-432	4.97	0.574	-0.427	-0.989	1.11
r	-6.23×10^{-4}	0.379	3.69	0	0	0	0	0
η	-0.277	2.27	465	-1.77	2.57	-1.03	-2.2	2.84
σ	0.557	-0.153	-71.4	5.74	2.96	0	0	0
$\varkappa$	0.211	1.79	336	-3.9	-0.73	0.35	0.928	-0.0322

Note: Entries are numerical partial derivatives, not elasticities. R_Δ is the ratio of average dollar illiquidation to average dollar liquidation. The money-response half-life $h_{1/2}$ is measured in days; adjustment frequencies and the average MPC are annualized. At each perturbation, we recompute the riskless policies and the asset-weighted occupation measure using the linear KFE approximation. The average MPC is the riskless occupation average. MATLAB generates the table using centered differences and checks smaller perturbations and finer grids.

Figure 9 complements the local derivatives with wider sweeps of four parameters, holding the remaining parameters at their baseline values (Table 2). Each row corresponds to one primitive parameter (ρ, π, η, $\varkappa$); each column to one outcome (m^*, T, $\overline{MPC}$, MPC at m^*). The dashed vertical line marks the baseline.

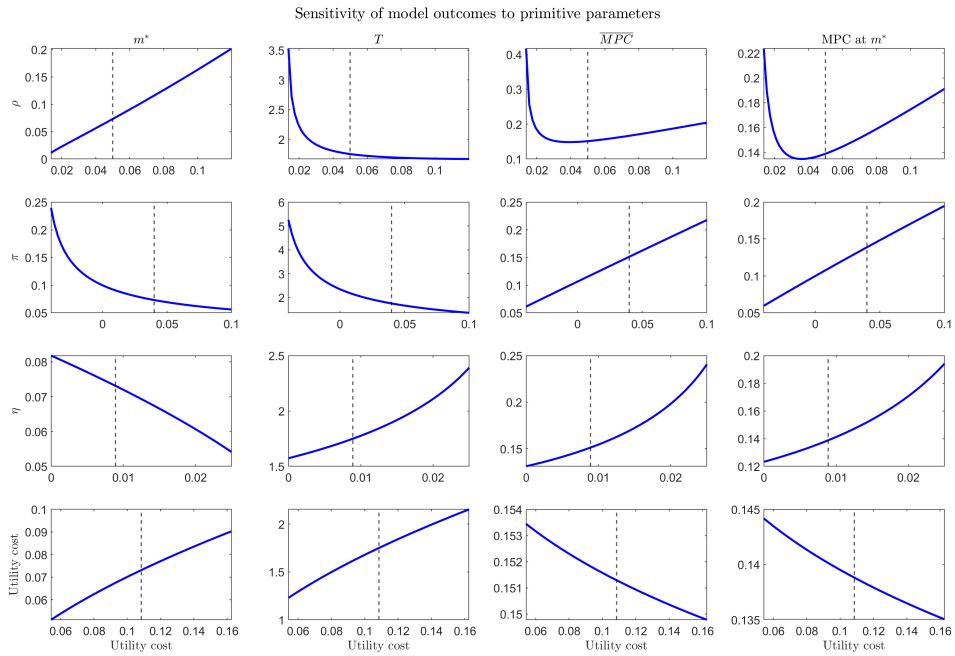


Figure 9: Sensitivity of key outcomes to primitive parameters. Rows: ρ (discount rate), π (inflation), η (liquid income share), $\varkappa$ (utility adjustment cost). Columns: optimal reset ratio m^*, time between resets T, average MPC $\overline{MPC}$, MPC at the reset point m^*. Dashed line: baseline calibration.